

Unit 1

PHYSICAL QUANTITIES AND UNITS

80 items: 24 from official papers (67 marks), 56 from the booklet

Syllabus sections in this unit

- 1.1 Physical quantities
- 1.2 SI units
- 1.3 Errors and uncertainties
- 1.4 Scalars and vectors

Unit 1: Physical quantities and units

1. M/J 26/P22/Q1/a

- 1 (a) By placing **one** tick (✓) in each row, complete Table 1.1 to identify whether each of the listed quantities is a scalar or a vector.

Table 1.1

quantity	scalar	vector
amplitude		
force		
momentum		
power		

[2]

2. M/J 26/P22/Q6/b(iii)

- (b) An α -particle travelling with a speed of $5.60 \times 10^6 \text{ m s}^{-1}$ collides with a stationary gold nucleus of mass 197 u. After the collision, the α -particle is travelling with a speed of $5.38 \times 10^6 \text{ m s}^{-1}$ in the opposite direction to its initial motion. No external forces act on this system.

- (iii) The absolute uncertainty in each of the measured speeds of the α -particle is $\pm 5 \times 10^4 \text{ m s}^{-1}$.

Calculate the percentage uncertainty in the change in velocity of the α -particle.

percentage uncertainty = % [2]

3. M/J 26/P23/Q4/b(i)

- 4 A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

- (b) The cell has an electromotive force (e.m.f.) of 4.90 V.

At a particular light intensity, the LDR has a resistance of 230Ω and a potential difference (p.d.) across it of 4.78 V.

- (i) Identify **one** possible source of systematic error in measuring the p.d. across the LDR using a digital voltmeter.

[1]

4. M/J 26/P24/Q1/a

- 1 (a) Underline **all** the vector quantities in the following list:

distance displacement speed velocity acceleration.

[1]

5. M/J 26/P24/Q2/b

also Unit 6 (3 marks); this unit 3 marks

- (b) In an experiment to determine the spring constant of a spring X, a student measures the original length of the spring. The student then suspends an object in equilibrium from the spring and measures the final length of the spring.

The spring does not extend beyond its limit of proportionality.

The measurements taken by the student are shown in Table 2.1.

Table 2.1

original length of spring X	$(2.1 \pm 0.1) \text{ cm}$
final length of spring X	$(14.0 \pm 0.1) \text{ cm}$
mass of suspended object	$(0.300 \pm 0.005) \text{ kg}$

Calculate:

- (i) the extension, in cm, of spring X together with its absolute uncertainty

extension = ($\pm$) cm [1]

- (ii) the spring constant of spring X

spring constant = N m^{-1} [3]

- (iii) the percentage uncertainty in the value of the spring constant calculated in **2(b)(ii)** due to the uncertainties in the measurements.

percentage uncertainty = % [2]

6. MAR 26/P22/Q1/b,c

- 1 (a) Define force.

[1]

- (b) Use the definition in **1(a)** to show that the SI base units of force are kg ms^{-2} .

[1]

- (c) A student is investigating the relationship between the force applied to the piston of a syringe and the velocity of fluid exiting the syringe. The student wishes to use the equation

$$v = \frac{Fk}{mA}$$

where:

v is the velocity of fluid exiting the syringe

F is the force applied to the piston

k is a constant with SI base units $\text{m}^2 \text{s}^{-1}$

m is the mass of the fluid in the syringe

A is the cross-sectional area of the piston.

By considering SI base units on both sides of the equation, state and explain whether this equation is homogeneous.

[2]

7. O/N 25/P21/Q2/b(ii)

- (b) A small steel ball of radius r and mass m falls vertically at terminal speed v through oil.

The viscous drag force D that acts on the ball is given by

$$D = 6\pi\eta rv$$

where η is a property of the oil called its viscosity.

- (ii) Determine the SI base units of η .

base units

[2]

8. O/N 25/P22/Q1/a,c

- 1 Scientists are investigating the variation in air pressure at different locations on a mountain.

- (a) The scientists take measurements of several physical quantities at each location.

Complete Table 1.1 by stating the SI base unit for each quantity and identifying with a tick (✓) whether each quantity is a scalar or a vector. Use the space for any working.

Table 1.1

quantity measured	SI base unit	scalar	vector
air temperature			
air pressure			

[2]

- (c) A quantity c relating to the motion of the balloon is calculated from three measured quantities k , F and v using the formula

$$c = \frac{2kF}{v^2}.$$

The percentage uncertainties in the measured quantities are given in Table 1.2.

Table 1.2

measured quantity	percentage uncertainty
k	5%
F	3%
v	4%

The calculated value of c is 1.8.

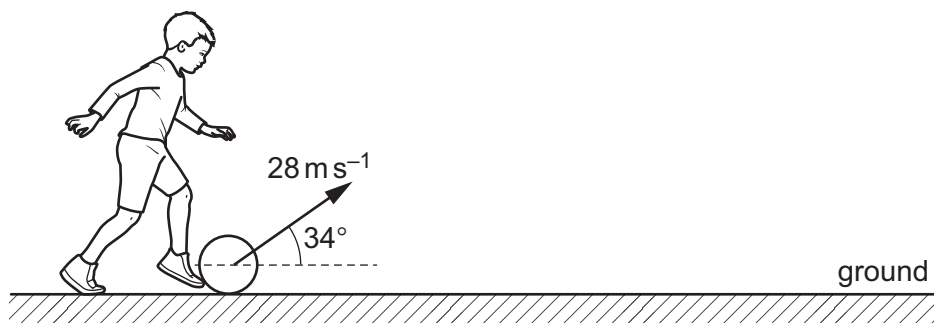
Determine the absolute uncertainty in c .

absolute uncertainty =

[2]

9. O/N 25/P24/Q1/a(i)

- 1 A child kicks a ball so that it leaves horizontal ground with a velocity of 28 m s^{-1} at an angle of 34° to the horizontal, as shown in Fig. 1.1.

**Fig. 1.1**

Air resistance is negligible. The ball leaves the ground at time $t = 0$.

- (a) (i) Calculate the horizontal component v_H and the vertical component v_V of the velocity of the ball immediately after it has left the ground.

$$v_H = \quad \text{ms}^{-1}$$

$$v_V = \quad \text{ms}^{-1}$$

[2]

10. O/N 25/P24/Q5/b(iii,iv)

- 5 A student uses a circuit containing an ammeter, a voltmeter and a cell to take measurements to determine the resistance of a length of nichrome wire.
- (b) The student also measures the length and the diameter of the wire. Table 5.1 shows the measurements recorded for each quantity.

Table 5.1

quantity	measurement
length	$(0.864 \pm 0.001) \text{ m}$
diameter	$(0.496 \pm 0.002) \text{ mm}$
voltmeter reading	$(1.38 \pm 0.02) \text{ V}$
ammeter reading	$(0.276 \pm 0.001) \text{ A}$

- (iii) Calculate the percentage uncertainty in ρ .

$$\text{percentage uncertainty} = \quad \% \quad [2]$$

- (iv) Determine the absolute uncertainty in ρ .

$$\text{absolute uncertainty} = \quad \Omega \text{ m} \quad [1]$$

11. M/J 25/P22/Q1/a

- 1 (a) Table 1.1 lists some physical quantities. Identify with ticks (✓) which quantities are vectors and which are scalars.

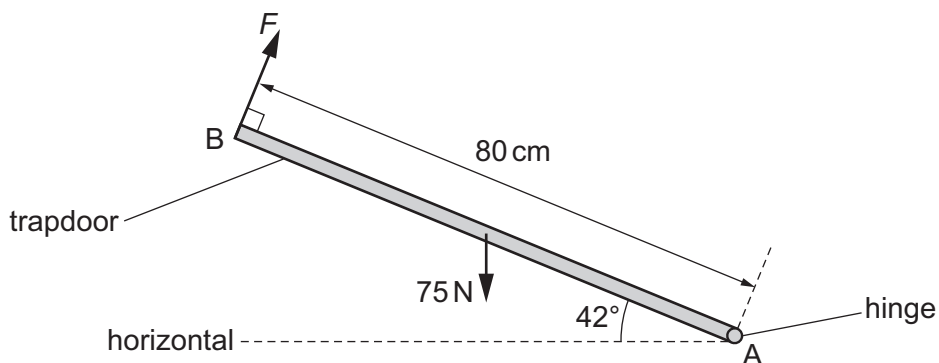
Table 1.1

quantity	scalar	vector
acceleration		
displacement		
gravitational potential energy		
speed		
temperature		

[2]

12. M/J 25/P24/Q1/b(ii)

- (b) A trapdoor has a hinge at end A, as shown in Fig. 1.1.

**Fig. 1.1** (not to scale)

The trapdoor has length 80 cm and weight 75 N. The mass of the trapdoor is uniformly distributed along its length.

A force F acts at right angles to the trapdoor at end B so that the trapdoor is held in equilibrium at an angle of 42° to the horizontal.

- (ii) Calculate the component of the weight that is perpendicular to the trapdoor.

component of weight =

N [1]

13. MAR 25/P22/Q1/a,b(ii,iii)

- 1** (a) Explain what is meant by the accuracy of a measured value.

[1]

- (b) Two solid cubes, A and B, are measured to determine the density of their materials.

Table 1.1 shows the measurements for cube A.

Table 1.1

quantity	measurement
length of side	$(1.53 \pm 0.01)\text{cm}$
mass	$(31.3 \pm 0.5)\text{g}$

- (ii) Calculate the percentage uncertainty in the density of the material of cube A.

percentage uncertainty = % [2]

- (iii) The density of the material of cube B is determined to be $9.2 \times 10^3 \text{ kg m}^{-3} \pm 6\%$.

State and explain whether cube A and cube B could be made from the same material.

[2]

14. O/N 24/P21/Q1/b(ii),c

(b) Fig. 1.1 shows a cuboidal glass block.

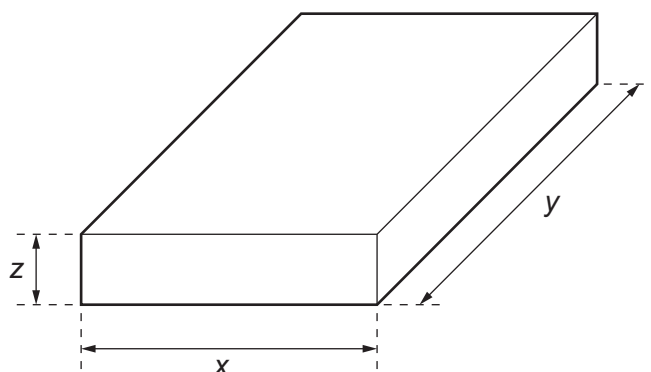


Fig. 1.1 (not to scale)

A student measures the mass m of the block and the side lengths x , y and z . The measurements are shown in Table 1.1.

Table 1.1

quantity	measurement
m	$(0.243 \pm 0.001) \text{ kg}$
x	$(5.41 \pm 0.01) \text{ cm}$
y	$(11.09 \pm 0.01) \text{ cm}$
z	$(1.62 \pm 0.01) \text{ cm}$

(i) Determine the density of the glass.

density = kg m^{-3} [2]

(ii) Calculate the percentage uncertainty in the density.

percentage uncertainty = % [3]

(c) The true value of the density of the glass is different from the answer in **(b)(i)** because of a systematic error in the measurements.

Suggest **one** possible cause of this systematic error.

[1]

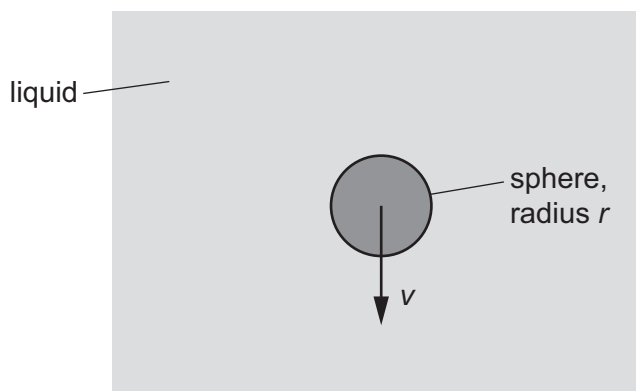
15. O/N 24/P22/Q1/a,b

also Unit 4 (2 marks), Unit 3 (1 mark); this unit 4 marks

- 1 (a) State what is meant by a vector quantity.

[1]

- (b) A sphere falls vertically through a liquid that has density 830 kg m^{-3} . The sphere has radius r and constant velocity v , as shown in Fig. 1.1.

**Fig. 1.1**

- (i) The drag force D acting on the sphere is given by

$$D = 6\pi r\eta v$$

where η is a property of the liquid.

Determine the SI base units of η .

SI base units

[3]

- (ii) State an equation showing the relationship between the magnitudes of the weight W , drag force D and upthrust U acting on the sphere.

[1]

- (iii) The volume of the sphere is 4.6 cm^3 . The drag force D is 0.32 N .

Calculate the weight of the sphere.

weight =

N [2]

16. M/J 24/P21/Q1/a

- 1 The drag force F_D acting on an object falling through air is given by

$$F_D = \frac{1}{2} C \rho A v^2$$

where A is the cross-sectional area of the object,

v is the velocity of the object in the air,

ρ is the density of the air and

C is a constant called the drag coefficient.

- (a) Use SI base units to show that the drag coefficient has no units.

[3]

17. M/J 24/P21/Q2/b(ii)

(b) A student throws a ball over a vertical wall of height h , as shown in Fig. 2.1.

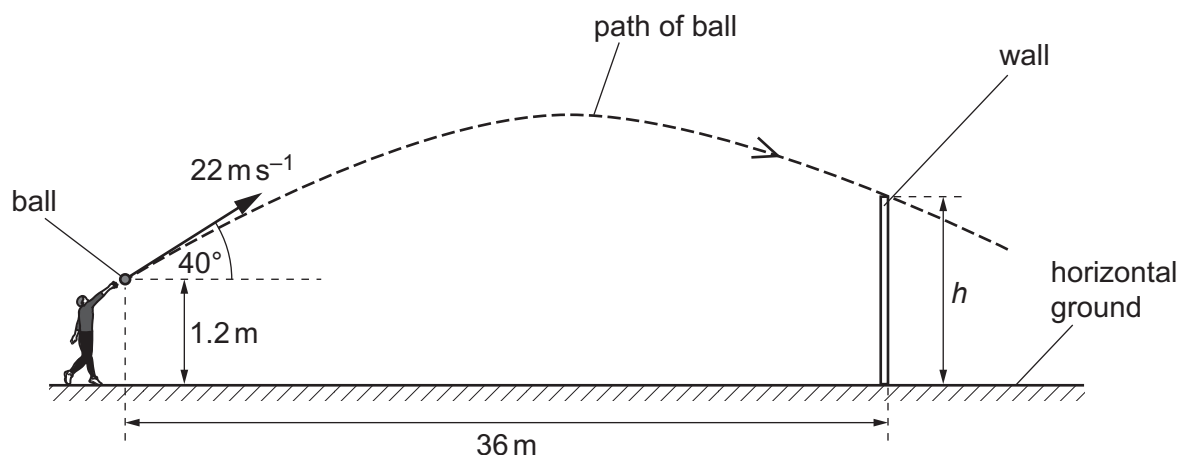


Fig. 2.1 (not to scale)

The ball leaves the hand of the student at a height of 1.2 m above the horizontal ground. The ball has an initial velocity of 22 m s^{-1} at an angle of 40° to the horizontal. The wall is a horizontal distance of 36 m from where the student releases the ball.

Air resistance is negligible.

(ii) Calculate the vertical component u of the initial velocity of the ball.

$$u = \quad \text{ms}^{-1} \quad [1]$$

18. M/J 24/P22/Q1/a

1 (a) The list below shows some SI quantities.

Underline the quantity that is **not** an SI base quantity.

charge current length time [1]

19. M/J 24/P23/Q1/a

1 The drag force F_D acting on a sphere falling through a liquid is given by

$$F_D = 6\pi\eta r v$$

where r is the radius of the sphere,
 v is the speed of the sphere in the liquid and
 η is a property of the liquid called the viscosity.

(a) Show that the SI base units of viscosity are $\text{kg m}^{-1} \text{s}^{-1}$.

[2]

20. MAR 24/P22/Q1/a,b

- 1 (a) Table 1.1 lists some SI quantities. Complete the table by indicating with a tick (✓) which rows are SI base quantities.

Table 1.1

quantity	base quantity
current	
energy	
force	
mass	

[1]

- (b) Use the definition of power to determine its SI base units.

SI base units

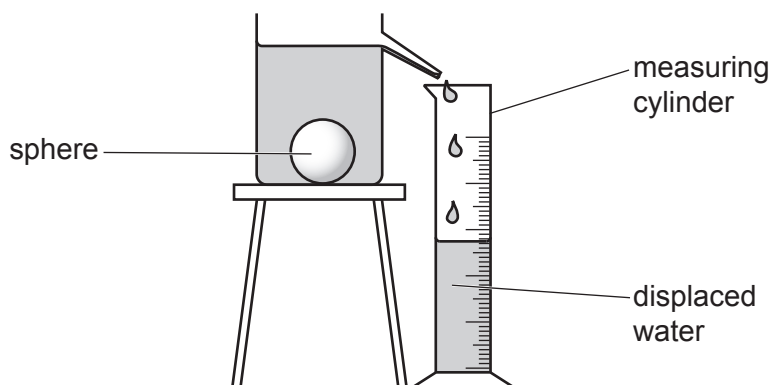
[2]

21. O/N 23/P21/Q1/a,b

- 1 (a) Compare scalar and vector quantities.

[2]

- (b) The radius of a small sphere is determined from a measurement of the volume of the sphere. The sphere is submerged in water, displacing some of the water into a measuring cylinder as shown in Fig. 1.1.

**Fig. 1.1** (not to scale)

The measured volume of displaced water is $(28.0 \pm 0.5) \text{ cm}^3$.

Calculate:

- (i) the radius, in cm, of the sphere

radius =

cm [1]

- (ii) the percentage uncertainty in the radius of the sphere.

percentage uncertainty =

% [2]

22. O/N 23/P22/Q1/a

- 1 (a) In the following list, underline **all** quantities that are SI base quantities.

charge electric current force time [1]

23. O/N 23/P23/Q1/a,b(i,iii)

- 1 (a) Table 1.1 lists some physical quantities. Complete the table by placing a tick (✓) next to the scalar quantities.

Table 1.1

acceleration	
charge	
momentum	
power	
upthrust	

[1]

- (b) A uniform cylinder has diameter D , length L and mass M .
The density ρ of the cylinder is given by

$$\rho = \frac{4M}{\pi D^2 L}.$$

Table 1.2 shows the data obtained from an experiment to determine the density of the cylinder.

Table 1.2

quantity	measurement	percentage uncertainty
D	$(26.2 \pm 0.1)\text{mm}$	%
L	$(162 \pm 1)\text{mm}$	%
M	$(247 \pm 1)\text{g}$	0.4%

- (i) Calculate the percentage uncertainties in D and L . Write your answers in Table 1.2.

[1]

- (iii) Calculate the percentage uncertainty in the density.

percentage uncertainty = % [2]

24. O/N 23/P23/Q2/a(ii)

- 2** A ball on horizontal ground is kicked towards a vertical wall. Fig. 2.1 shows the path of the ball.

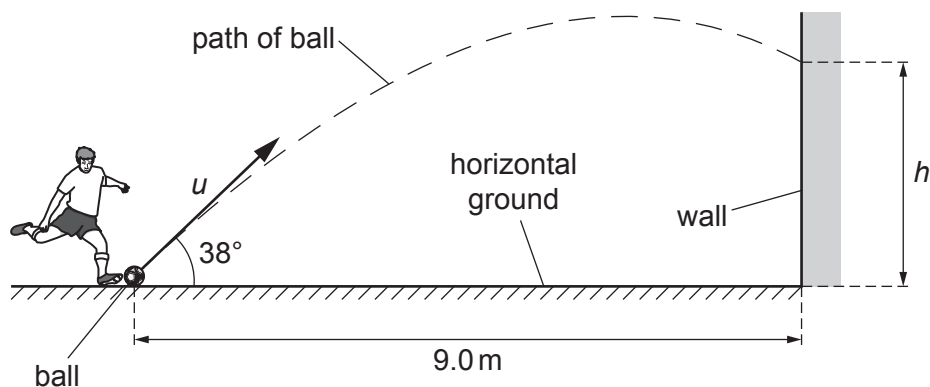


Fig. 2.1 (not to scale)

The ball has an initial velocity u at an angle of 38° to the ground. The ball travels a horizontal distance of 9.0 m before striking the wall at a height h above the ground. The horizontal component u_H of the initial velocity of the ball is 9.5 m s^{-1} .

Air resistance is negligible.

(a)

- (ii)** Calculate the vertical component u_V of the initial velocity of the ball.

$$u_V = \quad \text{ms}^{-1} \quad [2]$$

25. M/J 23/P22/Q1

booklet (scan + OCR)

(a) (i) Define pressure.

..... [1]

(ii) Use the answer to (a)(i) to show that the SI base units of pressure are $\text{kg m}^{-1} \text{s}^{-2}$.

[1]

(b) A horizontal pipe has length L and a circular cross-section of radius R . A liquid of density ρ flows through the pipe. The mass m of liquid flowing through the pipe in time t is given by

$$m = \frac{\pi(p_2 - p_1)R^4 \rho t}{8kL}$$

where p_1 and p_2 are the pressures at the ends of the pipe and k is a constant.Determine the SI base units of k .

SI base units [3]

(c) An experiment is performed to determine the value of k by measuring the values of the other quantities in the equation in (b).The values of L and R each have a percentage uncertainty of 2%.State and explain, quantitatively, which of these two quantities contributes more to the percentage uncertainty in the calculated value of k .

..... [1]

26. M/J 23/P21/Q1

booklet (scan + OCR)

(a) (i) Define power.

..... [1]

(ii) Use the definition of power to show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

[1]

(b) The intensity I of a sound wave moving through a gas is given by

$$I = f^2 A^2 v k$$

where f is the frequency of the wave, A is the amplitude of the wave, v is the speed of the waveand k is a constant that depends on the gas.Determine the SI base units of k .

SI base units [3]

27. MAR 23/P22/Q1

booklet (scan + OCR)

(a) Underline **all** the SI base units in the following list.

ampere coulomb current kelvin newton [1]

(b) A toy car moves in a horizontal straight line. The displacement s of the car is given by the equation

$$s = \frac{v^2}{2a}$$

where a is the acceleration of the car and v is its final velocity.State **two** conditions that apply to the motion of the car in order for the above equation to be valid.

1

2 [2]

(c) An experiment is performed to determine the acceleration of the car in **(b)**. The following measurements are obtained:

$$s = 3.89 \text{ m} \pm 0.5\%$$

$$v = 2.75 \text{ ms}^{-1} \pm 0.8\%$$

(i) Calculate the acceleration a of the car.

$$a = \dots \text{ ms}^{-2} \quad [1]$$

(ii) Determine the percentage uncertainty, to two significant figures, in a .

$$\text{percentage uncertainty} = \dots \% \quad [2]$$

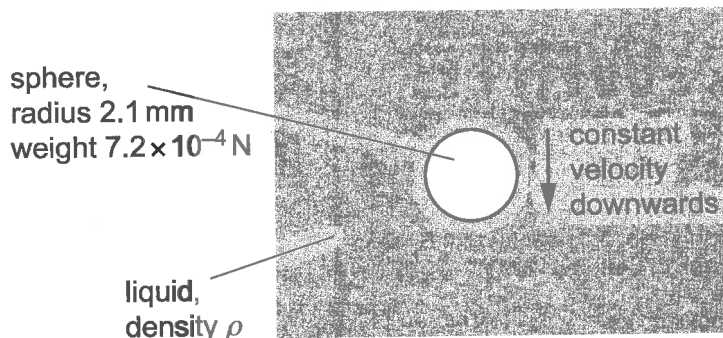
(iii) Use your answers in **(c)(i)** and **(c)(ii)** to determine the absolute uncertainty in the calculated value of a .

$$\text{absolute uncertainty} = \dots \text{ ms}^{-2} \quad [1]$$

28. MAR 22/P22/Q1

booklet (scan + OCR)

A sphere of radius 2.1 mm falls with terminal (constant) velocity through a liquid, as shown in Fig. 1.1.

**Fig. 1.1**

Three forces act on the moving sphere. The weight of the sphere is $7.2 \times 10^{-4} \text{ N}$ and the upthrust acting on it is $4.8 \times 10^{-4} \text{ N}$. The viscous force F_V acting on the sphere is given by

$$F_V = krv$$

where r is the radius of the sphere, v is its velocity and k is a constant. The value of k in SI units is 17.

(a) Determine the SI base units of k .

SI base units [2]

(b) Use the value of the upthrust acting on the sphere to calculate the density ρ of the liquid.

$\rho = \dots\dots\dots \text{ kg m}^{-3}$ [3]

(c) (i) On the sphere in Fig. 1.1, draw three arrows to show the directions of the weight W , the upthrust U and the viscous force F_V . Label these arrows W , U and F_V respectively. [1]

(ii) Determine the magnitude of the terminal (constant) velocity of the sphere.

velocity = ms^{-1} [2]

29. O/N 22/P22/Q1

booklet (scan + OCR)

(a) State what is meant by work done.

..... [1]

(b) Use the answer to **(a)** to determine the SI base units of power.

SI base units [2]

(c) The maximum useful output power P of a car travelling on a horizontal road is given by

$$P = v^3 b$$

where v is the maximum speed of the car and b is a constant.

For the car,

$$P = 84 \text{ kW} \pm 5\%$$

and $b = 0.56 \pm 7\%$ in SI units.**(i)** Calculate the value of v .

$$v = \dots \text{ ms}^{-1} [2]$$

(ii) Determine the absolute uncertainty in the value of v .

$$\text{absolute uncertainty} = \dots \text{ ms}^{-1} [2]$$

30. O/N 22/P21/Q1

booklet (scan + OCR)

- (a) The boxes in Fig. 1.1 contain terms on the left-hand side and examples of these terms on the right-hand side.

Draw a line between each term on the left and the correct example on the right.

base quantity	coulomb
base unit	electric current
derived quantity	force
derived unit	kilogram

Fig. 1.1**[2]**

- (b) A set of experimental measurements is described as precise and not accurate.
State what is meant by:

- (i) precise

..... [1]

- (ii) not accurate.

..... [1]

- (c) An object of mass m travels with speed v in a circle of radius r . The force F acting on the object is given by

$$F = \frac{mv^2}{r}.$$

The percentage uncertainties of three of the quantities are given in Table 1.1.

Table 1.1

quantity	percentage uncertainty
F	$\pm 3\%$
m	$\pm 4\%$
r	$\pm 5\%$

The value of v is determined from F , m and r .

- (i) Calculate the percentage uncertainty in v .

percentage uncertainty = % [2]

- (ii) The value of v is 15.0 ms^{-1} .

Calculate the absolute uncertainty in v .

absolute uncertainty = ms^{-1} [1]

31. O/N 22/P23/Q1

booklet (scan + OCR)

The rate of flow Q of a liquid along a narrow pipe of length L and radius r is given by

$$Q = \frac{\alpha r^4}{L}$$

where α is a constant.

An experiment is carried out to determine the value of α . The data from the experiment are shown in Table 1.1.

Table 1.1

quantity	value	percentage uncertainty
Q	$2.72 \times 10^{-8} \text{ m}^3 \text{ s}^{-1}$	$\pm 3\%$
r	$7.1 \times 10^{-5} \text{ m}$	$\pm 2\%$
L	$2.5 \times 10^{-2} \text{ m}$	$\pm 4\%$

- (a) Use information in Table 1.1 to show that the SI base unit of α is s^{-1} .

[1]

- (b) Show that the percentage uncertainty in α is 15%.

[1]

- (c) Calculate α with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

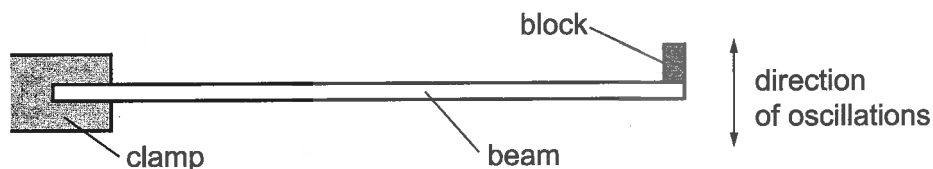
$$\alpha = (\dots\dots\dots \pm \dots\dots\dots) \times 10^7 \text{ s}^{-1} \quad [3]$$

32. M/J 22/P22/Q1

booklet (scan + OCR)

(a) In the following list, underline all units that are SI base units.

ampere degree Celsius kilogram newton [1]

(b) Fig. 1.1 shows a horizontal beam clamped at one end with a block attached to the other end.**Fig. 1.1**

The block is made to oscillate vertically.

The Young modulus E of the material of the beam is given by

$$E = \frac{kM}{T^2}$$

where M is the mass of the block, T is the period of the oscillationsand k is a constant.A student determines the values and percentage uncertainties of k , M and T .

Table 1.1 lists the percentage uncertainties.

Table 1.1

quantity	percentage uncertainty
k	$\pm 2.1\%$
M	$\pm 0.6\%$
T	$\pm 1.5\%$

The student uses the values of k , M and T to calculate the value of E as $8.245 \times 10^9 \text{ Pa}$.**(i)** Calculate the percentage uncertainty in the value of E .

percentage uncertainty = % [2]

(ii) Use your answer in **(b)(i)** to determine the value of E , with its absolute uncertainty, to an appropriate number of significant figures. $E = (\text{.....} \pm \text{.....}) \times 10^9 \text{ Pa}$ [2]

33. M/J 22/P23/Q1

booklet (scan + OCR)

A solid metal sphere has a diameter of (3.42 ± 0.02) cm and a mass of (67 ± 2) g.

(a) Calculate the density, in g cm^{-3} , of the metal.

density = g cm^{-3} [3]

(b) Determine the percentage uncertainty in the density.

percentage uncertainty = % [2]

34. O/N 21/P23/Q2

booklet (scan + OCR)

(a) State what is meant by *work done*.

..... [1]

(b) Use your answer in (a) to show that the SI base units of energy are $\text{kg m}^2 \text{s}^{-2}$.

[1]

(c) A metal rod is heated at one end so that thermal energy flows to the other end.

The thermal energy E that flows through the rod in time t is given by

$$E = \frac{cA(T_1 - T_2)t}{L}$$

where A is the cross-sectional area of the rod,

T_1 and T_2 are the temperatures of the ends of the rod,

L is the length of the rod,

and c is a constant.

Determine the SI base units of c .

SI base units [3]

35. O/N 21/P22/Q1

booklet (scan + OCR)

- (a) A unit may be stated with a prefix that represents a power-of-ten multiple or submultiple. Complete Table 1.1 to show the name and symbol of each prefix and the corresponding power-of-ten multiple or submultiple.

Table 1.1

prefix	power-of-ten multiple or submultiple
kilo (k)	10^3
tera (T)	
()	10^{-12}

[2]

- (b) In the following list, underline all the units that are SI base units.

ampere coulomb metre newton

[1]

- (c) The potential difference V between the two ends of a uniform metal wire is given by

$$V = \frac{4\rho LI}{\pi d^2}$$

where d is the diameter of the wire,

I is the current in the wire,

L is the length of the wire,

and ρ is the resistivity of the metal.

For a particular wire, the percentage uncertainties in the values of some of the above quantities are listed in Table 1.2.

Table 1.2

quantity	percentage uncertainty
d	$\pm 3.0\%$
I	$\pm 2.0\%$
L	$\pm 2.5\%$
V	$\pm 3.5\%$

The quantities listed in Table 1.2 have values that are used to calculate ρ as $4.1 \times 10^{-7} \Omega \text{ m}$.

For this value of ρ , calculate:

- (i) the percentage uncertainty

percentage uncertainty =% [2]

- (ii) the absolute uncertainty.

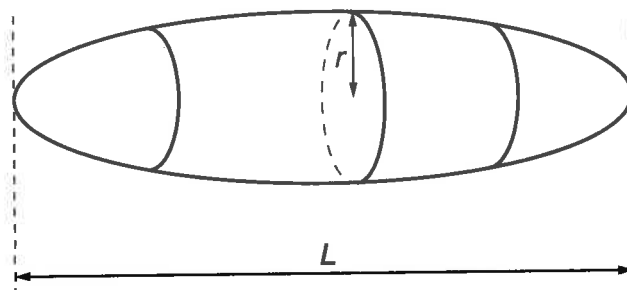
absolute uncertainty = $\Omega \text{ m}$ [1]

36. O/N 21/P21/Q1

booklet (scan + OCR)

(a) Define *density*.

..... [1]

(b) A smooth pebble, made from uniform rock, has the shape of an elongated sphere as shown in Fig. 1.1.**Fig. 1.1**

The length of the pebble is L . The cross-section of the pebble, in the plane perpendicular to L , is circular with a maximum radius r .

A student investigating the density of the rock makes measurements to determine the values of L , r and the mass M of the pebble as follows:

$$L = (0.1242 \pm 0.0001) \text{ m}$$

$$r = (0.0420 \pm 0.0004) \text{ m}$$

$$M = (1.072 \pm 0.001) \text{ kg}.$$

(i) State the name of a measuring instrument suitable for making this measurement of L .

..... [1]

(ii) Determine the percentage uncertainty in the measurement of r .

percentage uncertainty = % [1]

(c) The density ρ of the rock from which the pebble in (b) is composed is given by

$$\rho = \frac{Mr^n}{kL}$$

where n is an integer and k is a constant, with no units, that is equal to 2.094.

(i) Use SI base units to show that n is equal to -2 .

[2]

(ii) Calculate the percentage uncertainty in ρ .

percentage uncertainty = % [3]

(iii) Determine ρ with its absolute uncertainty. Give your values to the appropriate number of significant figures.

$\rho = (\dots\dots\dots \pm \dots\dots\dots) \text{ kg m}^{-3}$ [3]

37. M/J 21/P21/Q1

booklet (scan + OCR)

(a) Define *density*.

..... [1]

(b) Fig. 1.1 shows a solid pyramid with a square base.

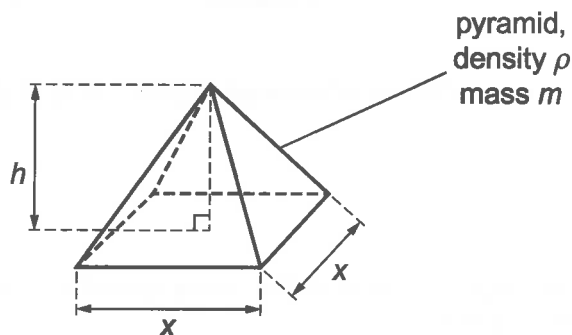


Fig. 1.1

The mass m of the pyramid is given by

$$m = \frac{1}{3}\rho hx^2$$

where ρ is the density of the material of the pyramid,
 h is the height, and
 x is the length of each side of the base.

Measurements are taken as shown in Table 1.1.

Table 1.1

quantity	measurement	percentage uncertainty
m	19.5g	$\pm 2\%$
x	4.0 cm	$\pm 5\%$
h	4.8 cm	$\pm 4\%$

- (i) Calculate the absolute uncertainty in length x .

absolute uncertainty = cm [1]

- (ii) The density ρ is calculated from the measurements in Table 1.1.
Determine the percentage uncertainty in the calculated value of ρ .

percentage uncertainty = % [2]

- (c) The square base of the pyramid in (b) rests on the horizontal surface of a bench.
Use data from Table 1.1 to calculate the average pressure of the pyramid on the surface of the bench. The uncertainty in your answer is not required.

pressure = Pa [3]

38. O/N 20/P23/Q1

booklet (scan + OCR)

(a) An electromagnetic wave has a wavelength of $85\text{ }\mu\text{m}$.

(i) State the wavelength, in m, of the wave.

wavelength = m [1]

(ii) Calculate the frequency, in THz, of the wave.

frequency = THz [2]

(iii) State the name of the region of the electromagnetic spectrum that contains this wave.

..... [1]

(b) The current I in a coil of wire produces a magnetic field. The energy E stored in the magnetic field is given by

$$E = \frac{I^2 L}{2}$$

where L is a constant.

The manufacturer of the coil states that the value of L , in SI base units, is $7.5 \times 10^{-6} \pm 5\%$.

The current I in the coil is measured as $(0.50 \pm 0.02)\text{ A}$.

The values of L and I are used to calculate E .

Determine the percentage uncertainty in the value of E .

percentage uncertainty = % [2]

39. M/J 20/P22/Q1

booklet (scan + OCR)

(a) Define *velocity*.

..... [1]

(b) The drag force F_D acting on a car moving with speed v along a straight horizontal road is given by

$$F_D = v^2 A k$$

where k is a constant and A is the cross-sectional area of the car.Determine the SI base units of k .

SI base units [2]

(c) The value of k , in SI base units, for the car in **(b)** is 0.24. The cross-sectional area A of the car is 5.1 m^2 .

The car is travelling with a constant speed along a straight road and the output power of the engine is $4.8 \times 10^4 \text{ W}$. Assume that the output power of the engine is equal to the rate at which the drag force F_D is doing work against the car.

Determine the speed of the car.

speed = ms^{-1} [3]

40. M/J 20/P21/Q1

booklet (scan + OCR)

- (a)** Use an expression for work done, in terms of force, to show that the SI base units of energy are $\text{kg m}^2 \text{s}^{-2}$.

[2]

- (b) (i)** The energy E stored in an electrical component is given by

$$E = \frac{Q^2}{2C}$$

where Q is charge and C is a constant.

Use this equation and the information in **(a)** to determine the SI base units of C .

SI base units [2]

- (ii)** Measurements of a constant current in a wire are taken using an analogue ammeter. For these measurements, describe one possible cause of:

1. a random error

.....

2. a systematic error.

..... [2]

41. MAR 20/P22/Q1

booklet (scan + OCR)

- (a) Length, mass and temperature are all SI base quantities.
State **two** other SI base quantities.

1.

2. [2]

- (b) The acceleration of free fall g may be determined from an oscillating pendulum using the equation

$$g = \frac{4\pi^2 l}{T^2}$$

where l is the length of the pendulum and T is the period of oscillation.

In an experiment, the measured values for an oscillating pendulum are

$$l = 1.50 \text{ m} \pm 2\%$$

and $T = 2.48 \text{ s} \pm 3\%$.

- (i) Calculate the acceleration of free fall g .

$$g = \dots \text{ ms}^{-2} \quad [1]$$

- (ii) Determine the percentage uncertainty in g .

$$\text{percentage uncertainty} = \dots \% \quad [2]$$

- (iii) Use your answers in (b)(i) and (b)(ii) to determine the absolute uncertainty of the calculated value of g .

$$\text{absolute uncertainty} = \dots \text{ ms}^{-2} \quad [1]$$

42. M/J 20/P23/Q1

booklet (scan + OCR)

- (a) State **one** similarity and **one** difference between *distance* and *displacement*.

similarity:

difference: [2]

- (b) A student takes several measurements of the same quantity. This set of measurements has high precision, but low accuracy.

Describe what is meant by:

- (i) *high precision*

..... [1]

- (ii) *low accuracy*.

..... [1]

43. O/N 19/P22/Q1

booklet (scan + OCR)

May be outside the 2025–27 syllabus

(a) Distinguish between vector and scalar quantities.

.....

..... [2]

(b) The electric field strength E at a distance x from an isolated point charge Q is given by the equation

$$E = \frac{Q}{x^2 b}$$

where b is a constant.**(i) Use the definition of electric field strength to show that E has SI base units of $\text{kg m A}^{-1} \text{s}^{-3}$.**

[2]

(ii) Use the units for E given in (b)(i) to determine the SI base units of b .SI base units of b [2]

44. O/N 19/P21/Q1

booklet (scan + OCR)

(a) Make estimates of:

- (i)**
- the mass, in g, of a new pencil

mass = g [1]

- (ii)**
- the wavelength of ultraviolet radiation.

wavelength = m [1]

- (b)**
- The period
- T
- of the oscillations of a mass
- m
- suspended from a spring is given by

$$T = 2\pi\sqrt{\frac{m}{k}}$$

where k is the spring constant of the spring.

The manufacturer of a spring states that it has a spring constant of $25 \text{ N m}^{-1} \pm 8\%$. A mass of $200 \times 10^{-3} \text{ kg} \pm 4 \times 10^{-3} \text{ kg}$ is suspended from the end of the spring and then made to oscillate.

- (i)**
- Calculate the period
- T
- of the oscillations.

 $T = \dots\dots\dots \text{ s}$ [1]

- (ii)**
- Determine the value of
- T
- , with its absolute uncertainty, to an appropriate number of significant figures.

 $T = \dots\dots\dots \pm \dots\dots\dots \text{ s}$ [3]

45. M/J 19/P21/Q1

booklet (scan + OCR)

(a) Define *velocity*.

..... [1]

(b) The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{kP}{\rho}}$$

where k is a constant that has no units.An experiment is performed to determine the value of k . The data from the experiment are shown in Fig. 1.1.

quantity	value	uncertainty
v	$3.3 \times 10^2 \text{ m s}^{-1}$	$\pm 3\%$
P	$9.9 \times 10^4 \text{ Pa}$	$\pm 2\%$
ρ	1.29 kg m^{-3}	$\pm 4\%$

Fig. 1.1**(i)** Use data from Fig. 1.1 to calculate k . $k =$ [2]**(ii)** Use your answer in **(b)(i)** and data from Fig. 1.1 to determine the value of k , with its absolute uncertainty, to an appropriate number of significant figures. $k =$ $\pm$ [3]

46. M/J 19/P23/Q1

booklet (scan + OCR)

(a) (i) Define *resistance*.

.....[1]

(ii) A potential difference of 0.60 V is applied across a resistor of resistance 4.0 GΩ.
Calculate the current, in pA, in the resistor.

current = pA [2]

(b) The energy E transferred when charge Q moves through an electrical component is given by the equation

$$E = QV$$

where V is the potential difference across the component.

Use the equation to determine the SI base units of potential difference.

SI base units [3]

47. MAR 19/P22/Q1

booklet (scan + OCR)

(a) The ampere, metre and second are SI base units.State **two** other SI base units.

1.

2. [2]

(b) The average drift speed v of electrons moving through a metal conductor is given by the equation:

$$v = \frac{\mu F}{e}$$

where e is the charge on an electron

F is a force acting on the electron

and μ is a constant.

Determine the SI base units of μ .

SI base units [3]

48. M/J 19/P22/Q1

booklet (scan + OCR)

- (a)**
- The diameter
- d
- of a cylinder is measured as
- $0.0125\text{ m} \pm 1.6\%$
- .

Calculate the absolute uncertainty in this measurement.

absolute uncertainty = m [1]

- (b)**
- The cylinder in
- (a)**
- stands on a horizontal surface. The pressure
- p
- exerted on the surface by the cylinder is given by

$$p = \frac{4W}{\pi d^2}.$$

The measured weight W of the cylinder is $0.38\text{ N} \pm 2.8\%$.

- (i)**
- Calculate the pressure
- p
- .

 $p = \dots\dots\dots \text{ N m}^{-2}$ [1]

- (ii)**
- Determine the absolute uncertainty in the value of
- p
- .

absolute uncertainty = N m^{-2} [2]**49. M/J 18/P22/Q1**

booklet (scan + OCR)

- (a)**
- Define
- force*
- .

.....[1]

- (b)**
- State the SI base units of force.

.....[1]

- (c)**
- The force
- F
- between two point charges is given by

$$F = \frac{Q_1 Q_2}{4\pi r^2 \epsilon}$$

where Q_1 and Q_2 are the charges, r is the distance between the charges, ϵ is a constant that depends on the medium between the charges.Use the above expression to determine the base units of ϵ .

base units[2]

50. M/J 18/P21/Q1

booklet (scan + OCR)

(a) State what is meant by a *scalar* quantity and by a *vector* quantity.

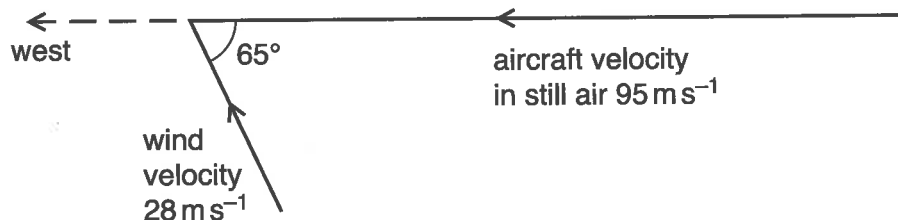
scalar:

vector: [2]

(b) Complete Fig. 1.1 to indicate whether each of the quantities is a vector or a scalar.

quantity	vector or scalar
power	
temperature	
momentum	

[2]

Fig. 1.1**(c)** An aircraft is travelling in wind. Fig. 1.2 shows the velocities for the aircraft in still air and for the wind.**Fig. 1.2**The velocity of the aircraft in still air is 95 m s^{-1} to the west.The velocity of the wind is 28 m s^{-1} from 65° south of east.**(i)** On Fig. 1.2, draw an arrow, labelled R, in the direction of the resultant velocity of the aircraft. [1]**(ii)** Determine the magnitude of the resultant velocity of the aircraft.magnitude of velocity = m s^{-1} [2]

51. M/J 18/P23/Q1

booklet (scan + OCR)

- (a)** An analogue voltmeter is used to take measurements of a constant potential difference across a resistor.

For these measurements, describe **one** example of

- (i)** a systematic error,

..... [1]

- (ii)** a random error.

..... [1]

- (b)** The potential difference across a resistor is measured as $5.0\text{ V} \pm 0.1\text{ V}$. The resistor is labelled as having a resistance of $125\ \Omega \pm 3\%$.

- (i)** Calculate the power dissipated by the resistor.

power = W [2]

- (ii)** Calculate the percentage uncertainty in the calculated power.

percentage uncertainty = % [2]

- (iii)** Determine the value of the power, with its absolute uncertainty, to an appropriate number of significant figures.

power = $\pm$ W [2]

52. O/N 17/P21/Q1

booklet (scan + OCR)

- (a)** The drag force F_D acting on a sphere moving through a fluid is given by the expression

$$F_D = K\rho v^2$$

where K is a constant,

ρ is the density of the fluid

and v is the speed of the sphere.

Determine the SI base units of K .

base units [3]

- (b)** A ball of weight 1.5 N falls vertically from rest in air. The drag force F_D acting on the ball is given by the expression in **(a)**. The ball reaches a constant (terminal) speed of 33 m s^{-1} .

Assume that the upthrust acting on the ball is negligible and that the density of the air is uniform.

For the instant when the ball is travelling at a speed of 25 m s^{-1} , determine

- (i)** the drag force F_D on the ball,

$F_D = \dots\dots\dots \text{ N}$ [2]

- (ii)** the acceleration of the ball.

acceleration = m s^{-2} [2]

- (c)** Describe the acceleration of the ball in **(b)** as its speed changes from zero to 33 m s^{-1} .

.....

 [3]

53. O/N 17/P23/Q1

booklet (scan + OCR)

(a) (i) Define *power*.

..... [1]

(ii) Show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

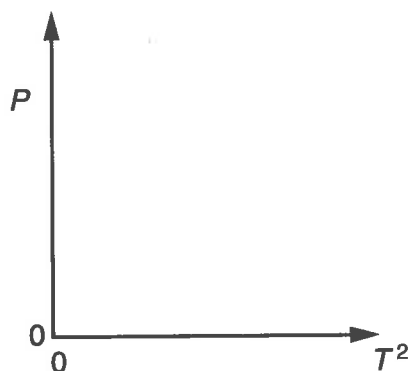
[1]

(b) All bodies radiate energy. The power P radiated by a body is given by

$$P = kAT^4$$

where T is the thermodynamic temperature of the body, A is the surface area of the bodyand k is a constant.**(i) Determine the SI base units of k .**

base units [2]

(ii) On Fig. 1.1, sketch the variation with T^2 of P . The quantity A remains constant.**Fig. 1.1**

[1]

54. M/J 17/P22/Q1

booklet (scan + OCR)

(a) State two SI base units other than kilogram, metre and second.

1.

2. [1]

(b) Determine the SI base units of resistivity.

base units [3]

(c) (i) A wire of cross-sectional area 1.5 mm^2 and length 2.5 m has a resistance of 0.030Ω .
Calculate the resistivity of the material of the wire in $\text{n}\Omega \text{ m}$.

resistivity = $\text{n}\Omega \text{ m}$ [3]

(ii) 1. State what is meant by *precision*.

.....

2. Explain why the precision in the value of the resistivity is improved by using a micrometer screw gauge rather than a metre rule to measure the diameter of the wire.

.....

..... [2]

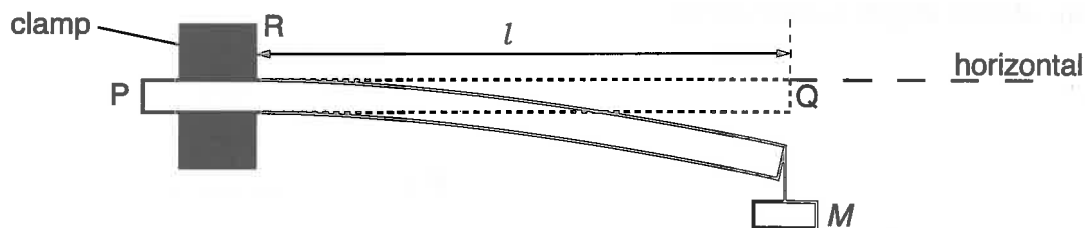
55. M/J 17/P21/Q1

booklet (scan + OCR)

(a) Determine the SI base units of stress.

Show your working.

base units [2]

(b) A beam PQ is clamped so that the beam is horizontal. A mass M of 500g is hung from end Q and the beam bends slightly, as illustrated in Fig. 1.1.**Fig. 1.1**

The length l of the beam from the edge of the clamp R to end Q is 60.0 cm. The width b of the beam is 30.0 mm and the thickness d of the beam is 5.00 mm. The material of the beam has Young modulus E .

The mass M is made to oscillate vertically. The time period T of the oscillations is 0.58 s.

The period T is given by the expression

$$T = 2\pi \sqrt{\frac{4Ml^3}{Ebd^3}}$$

(i) Determine E in GPa. $E = \dots\dots\dots$ GPa [3]**(ii) The quantities used to determine E should be measured with accuracy and with precision.**

1. Explain the difference between accuracy and precision.

accuracy:

precision: [2]

2. In a particular experiment, the quantities l and T are measured with the same percentage uncertainty. State and explain which of these two quantities contributes more to the uncertainty in the value of E .

..... [1]

56. M/J 17/P23/Q1

booklet (scan + OCR)

(a) Two forces, with magnitudes 5.0 N and 12 N, act from the same point on an object.

Calculate the magnitude of the resultant force R for the forces acting

(i) in opposite directions,

$$R = \dots\dots\dots \text{N [1]}$$

(ii) at right angles to each other.

$$R = \dots\dots\dots \text{N [1]}$$

- (b)** An object X rests on a smooth horizontal surface. Two horizontal forces act on X as shown in Fig. 1.1.

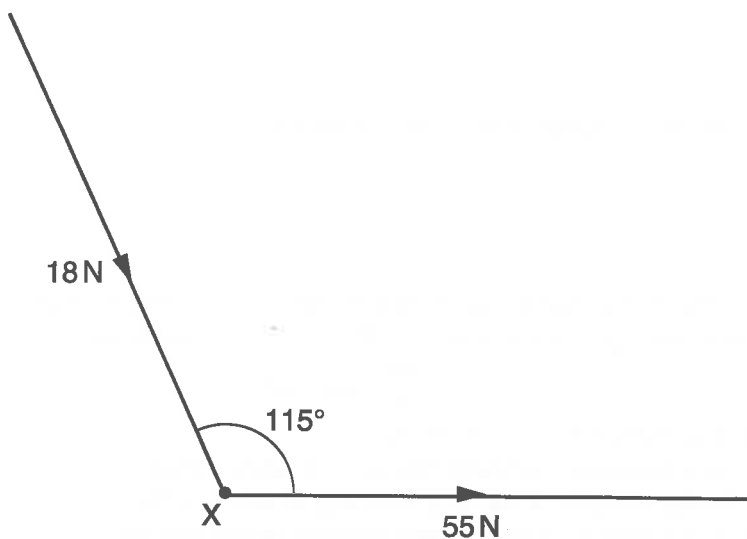


Fig. 1.1 (not to scale)

A force of 55 N is applied to the right. A force of 18 N is applied at an angle of 115° to the direction of the 55 N force.

- (i)** Use the resolution of forces or a scale diagram to show that the magnitude of the resultant force acting on X is 65 N.

[2]

- (ii)** Determine the angle between the resultant force and the 55 N force.

angle = $^\circ$ [2]

- (c)** A third force of 80 N is now applied to X in the opposite direction to the resultant force in **(b)**.

The mass of X is 2.7 kg.

Calculate the magnitude of the acceleration of X.

acceleration = ms^{-2} [3]

57. O/N 17/P22/Q1

booklet (scan + OCR)

One end of a wire is connected to a fixed point. A load is attached to the other end so that the wire hangs vertically.

The diameter d of the wire and the load F are measured as

$$d = 0.40 \pm 0.02 \text{ mm},$$

$$F = 25.0 \pm 0.5 \text{ N}.$$

(a) For the measurement of the diameter of the wire, state

(i) the name of a suitable measuring instrument,

..... [1]

(ii) how random errors may be reduced when using the instrument in **(i)**.

.....
..... [2]

(b) The stress σ in the wire is calculated by using the expression

$$\sigma = \frac{4F}{\pi d^2}.$$

(i) Show that the value of σ is $1.99 \times 10^8 \text{ N m}^{-2}$.

[1]

(ii) Determine the percentage uncertainty in σ .

percentage uncertainty =% [2]

(iii) Use the information in **(b)(i)** and your answer in **(b)(ii)** to determine the value of σ , with its absolute uncertainty, to an appropriate number of significant figures.

$\sigma = \dots\dots\dots \pm \dots\dots\dots \text{ N m}^{-2}$ [2]

58. O/N 16/P22/Q1

booklet (scan + OCR)

(a) (i) Define *pressure*.

..... [1]

(ii) Show that the SI base units of pressure are $\text{kg m}^{-1} \text{s}^{-2}$.

[1]

(b) Gas flows through the narrow end (nozzle) of a pipe. Under certain conditions, the mass m of gas that flows through the nozzle in a short time t is given by

$$\frac{m}{t} = kC\sqrt{\rho P}$$

where k is a constant with no units, C is a quantity that depends on the nozzle size, ρ is the density of the gas arriving at the nozzle, P is the pressure of the gas arriving at the nozzle.Determine the base units of C .

base units[3]

59. M/J 16/P23/Q1

booklet (scan + OCR)

- (a) A list of quantities that are either scalars or vectors is shown in Fig. 1.1.

quantity	scalar	vector
distance	✓	
energy		
momentum		
power		
time		
weight		

Fig. 1.1

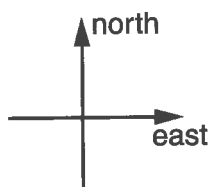
Complete Fig. 1.1 to indicate whether each quantity is a scalar or a vector.

One line has been completed as an example.

[2]

- (b) A girl runs 120 m due north in 15 s. She then runs 80 m due east in 12 s.

- (i) Sketch a vector diagram to show the path taken by the girl. Draw and label her resultant displacement R.



[1]

- (ii) Calculate, for the girl,

1. the average speed,

average speed = ms^{-1} [1]

2. the magnitude of the average velocity v and its angle with respect to the direction of the initial path.

magnitude of v = ms^{-1}

angle = $^{\circ}$ [3]

60. O/N 16/P21/Q1

booklet (scan + OCR)

(a) Define *density*.

.....[1]

(b) The mass m of a metal sphere is given by the expression

$$m = \frac{\pi d^3 \rho}{6}$$

where ρ is the density of the metal and d is the diameter of the sphere.
Data for the density and the mass are given in Fig. 1.1.

quantity	value	uncertainty
ρ	8100 kg m ⁻³	± 5%
m	7.5 kg	± 4%

Fig. 1.1**(i) Calculate the diameter d .** $d = \dots\dots\dots$ m [1]

(ii) Use your answer in (i) and the data in Fig. 1.1 to determine the value of d , with its absolute uncertainty, to an appropriate number of significant figures.

 $d = \dots\dots\dots \pm \dots\dots\dots$ m [3]

61. M/J 16/P21/Q1

booklet (scan + OCR)

(a) Make estimates of

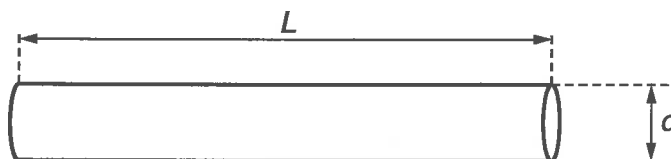
- (i)**
- the mass, in kg, of a wooden metre rule,

mass = kg [1]

- (ii)**
- the volume, in cm
- ³
- , of a cricket ball or a tennis ball.

volume = cm³ [1]

- (b)**
- A metal wire of length
- L
- has a circular cross-section of diameter
- d
- , as shown in Fig. 1.1.

**Fig. 1.1**The volume V of the wire is given by the expression

$$V = \frac{\pi d^2 L}{4}.$$

The diameter, length and mass M are measured to determine the density of the metal of the wire. The measured values are:

$$d = 0.38 \pm 0.01 \text{ mm},$$

$$L = 25.0 \pm 0.1 \text{ cm},$$

$$M = 0.225 \pm 0.001 \text{ g}.$$

Calculate the density of the metal, with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

density = kg m^{-3} [5]

62. M/J 16/P23/Q2

booklet (scan + OCR)

- (a)** Describe the effects, one in each case, of systematic errors and random errors when using a micrometer screw gauge to take readings for the diameter of a wire.

systematic errors:

random errors: [2]

- (b)** Distinguish between precision and accuracy when measuring the diameter of a wire.

precision:

accuracy: [2]

63. O/N 15/P22/Q1

booklet (scan + OCR)

- (a) The frequency of an X-ray wave is 4.6×10^{20} Hz.
Calculate the wavelength in pm.

wavelength = pm [3]

- (b) The distance from Earth to a star is 8.5×10^{16} m. Calculate the time for light to travel from the star to Earth in Gs.

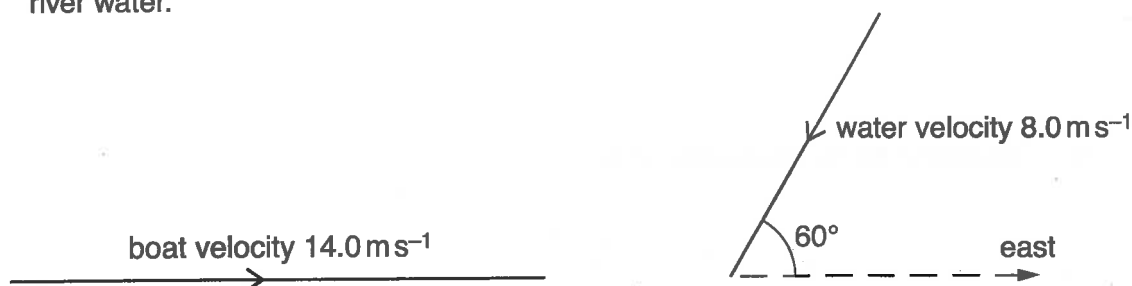
time = Gs [2]

- (c) The following list contains scalar and vector quantities.

Underline **all** the scalar quantities.

acceleration force mass power temperature weight [1]

- (d) A boat is travelling in a flowing river. Fig. 1.1 shows the velocity vectors for the boat and the river water.

**Fig. 1.1**

The velocity of the boat in still water is 14.0 ms^{-1} to the east. The velocity of the water is 8.0 ms^{-1} from 60° north of east.

- (i) On Fig. 1.1, draw an arrow to show the direction of the resultant velocity of the boat. [1]
(ii) Determine the magnitude of the resultant velocity of the boat.

magnitude of velocity = ms^{-1} [2]

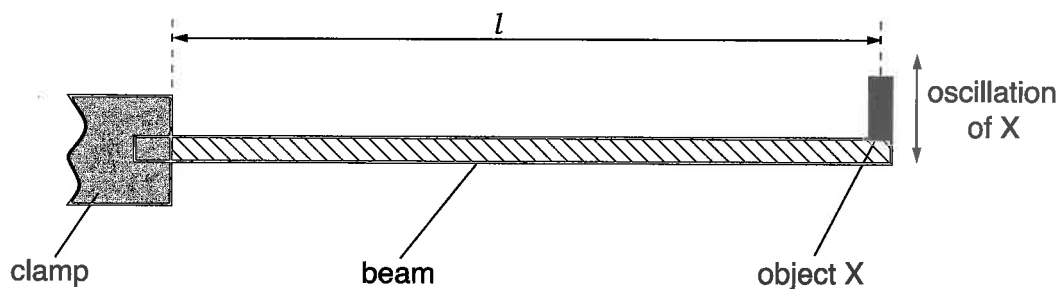
64. O/N 15/P21/Q1

booklet (scan + OCR)

(a) State two SI base quantities other than mass, length and time.

1.

2. [2]

(b) A beam is clamped at one end and an object X is attached to the other end of the beam, as shown in Fig. 1.1.**Fig. 1.1**

The object X is made to oscillate vertically.

The time period T of the oscillations is given by

$$T = K \sqrt{\frac{Ml^3}{E}}$$

where M is the mass of X, l is the length between the clamp and X, E is the Young modulus of the material of the beamand K is a constant.**(i)** 1. Show that the SI base units of the Young modulus are $\text{kg m}^{-1} \text{s}^{-2}$.2. Determine the SI base units of K .

[1]

SI base units of K [2]

(ii) Data in SI units for the oscillations of X are shown in Fig. 1.2.

quantity	value	uncertainty
T	0.45	$\pm 2.0\%$
l	0.892	$\pm 0.2\%$
M	0.2068	$\pm 0.1\%$
K	1.48×10^5	$\pm 1.5\%$

Fig. 1.2

Calculate E and its actual uncertainty.

$$E = \dots\dots\dots \pm \dots\dots\dots \text{ kg m}^{-1} \text{ s}^{-2} \text{ [4]}$$

65. O/N 15/P23/Q1

booklet (scan + OCR)

(a) The intensity of a progressive wave is defined as the average power transmitted through a surface per unit area.

Show that the SI base units of intensity are kg s^{-3} .

[2]

(b) (i) The intensity I of a sound wave is related to the amplitude x_0 of the wave by

$$I = K\rho cf^2x_0^2$$

where ρ is the density of the medium through which the sound is passing,

c is the speed of the sound wave, f is the frequency of the sound wave

and K is a constant.

Show that K has no units.

[2]

(ii) Calculate the intensity, in pW m^{-2} , of a sound wave where

$$K = 20,$$

$$\rho = 1.2 \text{ in SI base units,}$$

$$c = 330 \text{ in SI base units,}$$

$$f = 260 \text{ in SI base units and } x_0 = 0.24 \text{ nm.}$$

$$\text{intensity} = \dots\dots\dots \text{ pW m}^{-2} \text{ [3]}$$

66. M/J 15/P22/Q1

booklet (scan + OCR)

(a) Use the definition of work done to show that the SI base units of energy are $\text{kg m}^2 \text{s}^{-2}$.

[2]

(b) Define potential difference.

..... [1]

(c) Determine the SI base units of resistance. Show your working.

units [3]

67. M/J 15/P21/Q1

booklet (scan + OCR)

(a) Use the definition of power to show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

[2]

(b) Use an expression for electrical power to determine the SI base units of potential difference.

units [2]

68. M/J 15/P23/Q1

booklet (scan + OCR)

(a) The distance between the Sun and the Earth is 1.5×10^{11} m. State this distance in Gm.

distance = Gm [1]

(b) The distance from the centre of the Earth to a satellite above the equator is 42.3 Mm. The radius of the Earth is 6380 km.

A microwave signal is sent from a point on the Earth directly below the satellite.

Calculate the time taken for the microwave signal to travel to the satellite and back.

time = s [2]

- (c) The speed v of a sound wave through a gas of density ρ and pressure P is given by

$$v = \sqrt{\frac{CP}{\rho}}$$

where C is a constant.

Show that C has no unit.

[3]

- (d) Underline all the scalar quantities in the list below.

acceleration energy momentum power weight

[1]

- (e) A boat travels across a river in which the water is moving at a speed of 1.8 m s^{-1} .
The velocity vectors for the boat and the river water are shown to scale in Fig. 1.1.

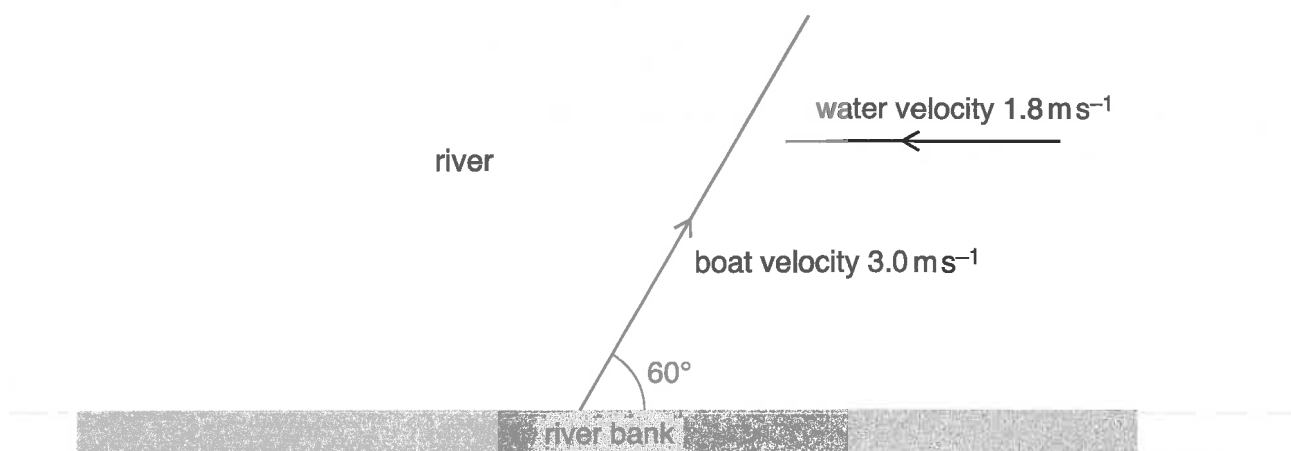


Fig. 1.1 (shown to scale)

In still water the speed of the boat is 3.0 m s^{-1} . The boat is directed at an angle of 60° to the river bank.

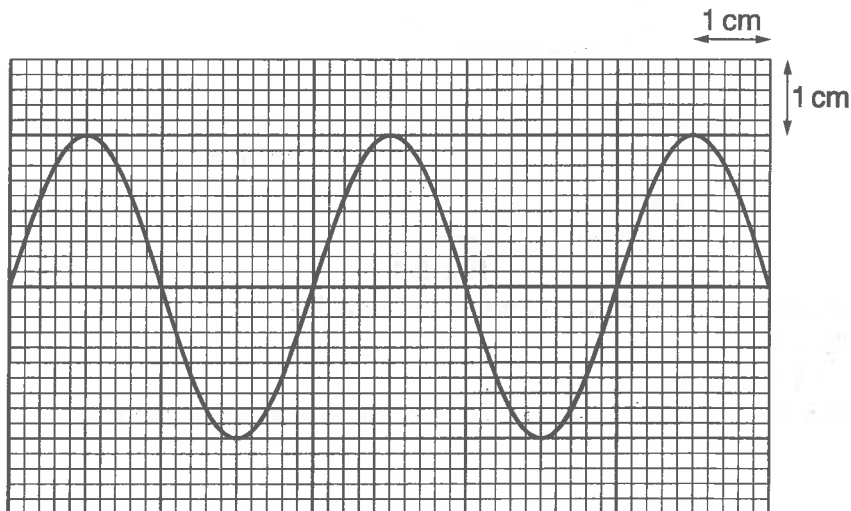
- (i) On Fig. 1.1, draw a vector triangle or a scale diagram to show the resultant velocity of the boat. [2]
- (ii) Determine the magnitude of the resultant velocity of the boat.

resultant velocity = m s^{-1} [2]

69. O/N 15/P21/Q2

booklet (scan + OCR)

The signal from a microwave detector is recorded on a cathode-ray oscilloscope (c.r.o.), as shown in Fig. 2.1.

**Fig. 2.1**

The time-base setting on the c.r.o. is 50 pscm^{-1} .

(a) Using Fig. 2.1, determine the wavelength of the microwaves.

wavelength = m [4]

(b) The signal from a radio wave detector is recorded on the same c.r.o.

The wavelength of the radio waves is $1.5 \times 10^3 \text{ m}$.

Determine the time-base setting required to display the same number of oscillations on the c.r.o. as shown in Fig. 2.1.

time-base setting = unit..... [2]

70. O/N 14/P21/Q1

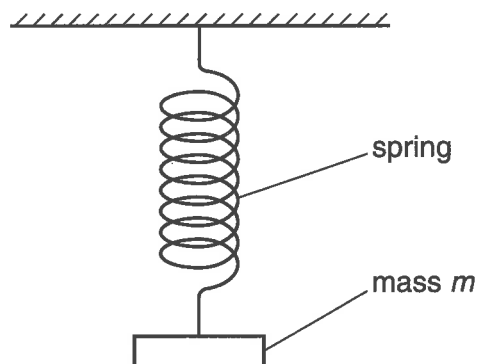
booklet (scan + OCR)

(a) Mass, length and time are SI base quantities.

State two other base quantities.

1.

2. [2]

(b) A mass m is placed on the end of a spring that is hanging vertically, as shown in Fig. 1.1.**Fig. 1.1**The mass is made to oscillate vertically. The time period of the oscillations of the mass is T .The period T is given by

$$T = C \sqrt{\frac{m}{k}}$$

where C is a constant and k is the spring constant.Show that C has no units.

[3]

71. O/N 14/P23/Q1

booklet (scan + OCR)

(a) The kilogram, metre and second are SI base units.

State two other base units.

1.

2. [2]

(b) Determine the SI base units of**(i)** stress,

SI base units[2]

(ii) the Young modulus.

SI base units[1]

72. M/J 14/P22/Q1

booklet (scan + OCR)

(a) Show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

[3]

(b) The rate of flow of thermal energy $\frac{Q}{t}$ in a material is given by

$$\frac{Q}{t} = \frac{CA\Delta T}{x}$$

where A is the cross-sectional area of the material, ΔT is the temperature difference across the thickness of the material, x is the thickness of the material, C is a constant.Determine the SI base units of C .

base units [4]

73. M/J 14/P23/Q1

booklet (scan + OCR)

(a) Underline **all** the base quantities in the following list.

ampere charge current mass second temperature weight [2]

(b) The potential energy E_p stored in a stretched wire is given by

$$E_p = \frac{1}{2}C\sigma^2V$$

where C is a constant, σ is the strain, V is the volume of the wire.Determine the SI base units of C .

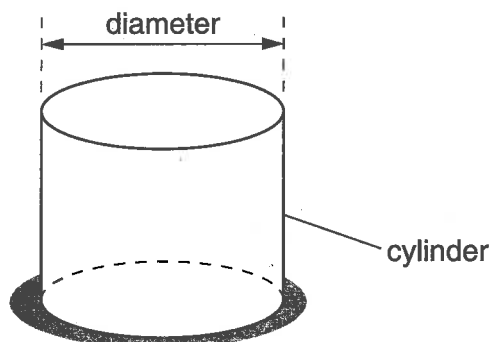
base units [3]

74. O/N 14/P21/Q2

booklet (scan + OCR)

(a) Define *pressure*.

..... [1]

(b) A cylinder is placed on a horizontal surface, as shown in Fig. 2.1.**Fig. 2.1**

The following measurements were made on the cylinder:

mass = 5.09 ± 0.01 kgdiameter = 9.4 ± 0.1 cm.**(i) Calculate the pressure produced by the cylinder on the surface.**

pressure = Pa [3]

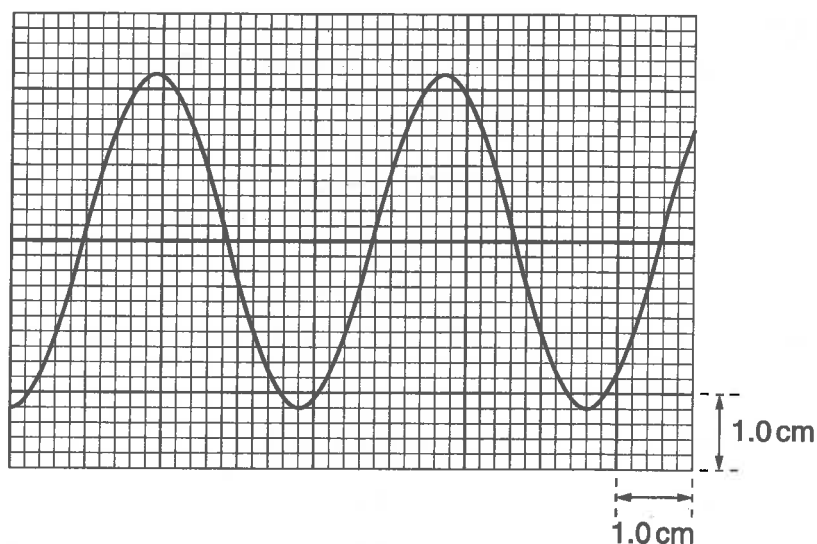
(ii) Calculate the actual uncertainty in the pressure.

actual uncertainty = Pa [3]

(iii) State the pressure, with its actual uncertainty.pressure = $\pm$ Pa [1]**75. O/N 14/P23/Q2**

booklet (scan + OCR)

A microphone detects a musical note of frequency f . The microphone is connected to a cathode-ray oscilloscope (c.r.o.). The signal from the microphone is observed on the c.r.o. as illustrated in Fig. 2.1.

**Fig. 2.1**

The time-base setting of the c.r.o. is 0.50 ms cm^{-1} . The Y-plate setting is 2.5 mV cm^{-1} .

(a) Use Fig. 2.1 to determine

(i) the amplitude of the signal,

amplitude = mV [2]

(ii) the frequency f ,

f = Hz [3]

(iii) the actual uncertainty in f caused by reading the scale on the c.r.o.

actual uncertainty = Hz [2]

(b) State f with its actual uncertainty.

f = $\pm$ Hz [1]

76. M/J 14/P22/Q2

booklet (scan + OCR)

A coin is made in the shape of a thin cylinder, as shown in Fig. 2.1.

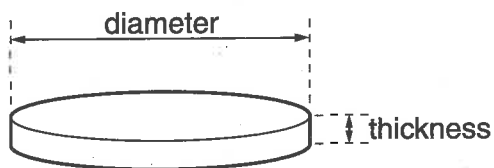
**Fig. 2.1**

Fig. 2.2 shows the measurements made in order to determine the density ρ of the material used to make the coin.

quantity	measurement	uncertainty
mass	9.6 g	± 0.5 g
thickness	2.00 mm	± 0.01 mm
diameter	22.1 mm	± 0.1 mm

Fig. 2.2

(a) Calculate the density ρ in kg m^{-3} .

$$\rho = \dots\dots\dots \text{kg m}^{-3} \quad [3]$$

(b) (i) Calculate the percentage uncertainty in ρ .

$$\text{percentage uncertainty} = \dots\dots\dots [3]$$

(ii) State the value of ρ with its actual uncertainty.

$$\rho = \dots\dots\dots \pm \dots\dots\dots \text{kg m}^{-3} \quad [1]$$

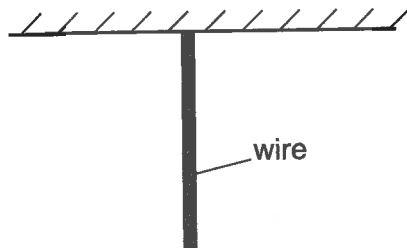
77. O/N 13/P21/Q1

booklet (scan + OCR)

(a) State two SI base units other than the kilogram, metre and second.

1.

2. [2]

(b) A metal wire has original length l_0 . It is then suspended and hangs vertically as shown in Fig. 1.1.**Fig. 1.1**

The weight of the wire causes it to stretch. The elastic potential energy stored in the wire is E .

(i) Show that the SI base units of E are $\text{kg m}^2 \text{s}^{-2}$.

[2]

(ii) The elastic potential energy E is given by

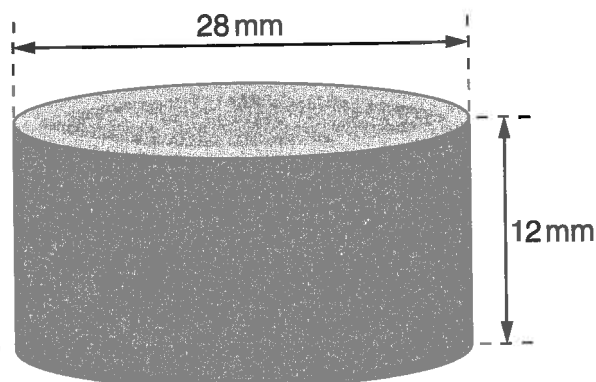
$$E = C\rho^2 g^2 A l_0^3$$

where ρ is the density of the metal, g is the acceleration of free fall, A is the cross-sectional area of the wireand C is a constant.Determine the SI base units of C .SI base units of C [3]

78. O/N 13/P23/Q1

booklet (scan + OCR)

A cylindrical disc is shown in Fig. 1.1.

**Fig. 1.1**

The disc has diameter 28 mm and thickness 12 mm.

The material of the disc has density $6.8 \times 10^3 \text{ kg m}^{-3}$.

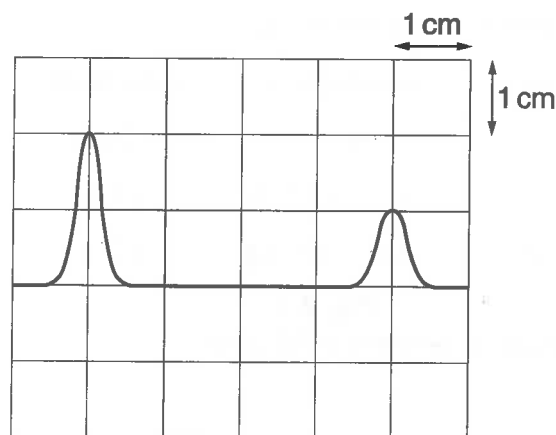
Calculate, to two significant figures, the weight of the disc.

weight = N [4]

79. O/N 13/P21/Q2

booklet (scan + OCR)

A source of radio waves sends a pulse towards a reflector. The pulse returns from the reflector and is detected at the same point as the source. The emitted and reflected pulses are recorded on a cathode-ray oscilloscope (c.r.o.) as shown in Fig. 2.1.

**Fig. 2.1**

The time-base setting is $0.20 \mu\text{s cm}^{-1}$.

(a) Using Fig. 2.1, determine the distance between the source and the reflector.

distance = m [4]

(b) Determine the time-base setting required to produce the same separation of pulses on the c.r.o. when sound waves are used instead of radio waves.

The speed of sound is 300 m s^{-1} .

.....

[3]

80. O/N 13/P23/Q2

booklet (scan + OCR)

The time T for a satellite to orbit the Earth is given by

$$T = \sqrt{\left(\frac{KR^3}{M}\right)}$$

where R is the distance of the satellite from the centre of the Earth,

M is the mass of the Earth,

and K is a constant.

(a) Determine the SI base units of K .

SI base units of K [2]

(b) Data for a particular satellite are given in Fig. 2.1.

quantity	measurement	uncertainty
T	$8.64 \times 10^4 \text{ s}$	$\pm 0.5\%$
R	$4.23 \times 10^7 \text{ m}$	$\pm 1\%$
M	$6.0 \times 10^{24} \text{ kg}$	$\pm 2\%$

Fig. 2.1

Calculate K and its actual uncertainty in SI units.

$K = \dots \pm \dots$ SI units [4]

Answers Section

1. M/J 26/P22/Q1/a

1(a)	amplitude and power identified as scalars	B1	Allow methods for marking the table other than a tick (e.g. ×) provided no ticks are seen
	force and momentum identified as vectors	B1	

2. M/J 26/P22/Q6/b(iii)

6(b)(iii)	(absolute uncertainty) $= 5 \times 10^4 + 5 \times 10^4$ $= 1 \times 10^5$	C1	Allow $2 \times 5 \times 10^4 = 1 \times 10^5$ Allow Δv to mean absolute uncertainty in v
	percentage uncertainty = $(1 \times 10^5 / (5.60 \times 10^6 + 5.38 \times 10^6)) \times 100$ $= 0.9\%$	A1	Allow answers to 2 or more SF that round to 0.9

3. M/J 26/P23/Q4/b(i)

4(b)(i)	zero error / calibration error	B1	Allow 'contact resistance error in the connections'. Allow 'resistance of connecting wires'. Allow 'not calibrated' BOD Apply List Rule
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4. M/J 26/P24/Q1/a

1(a)	displacement, velocity and acceleration underlined only.	A1	Allow other methods of indicating such as circling, ticking, but not crossing through. All three need to be indicated by the same notation
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5. M/J 26/P24/Q2/b

2(b)(i)	$(11.9 \pm 0.2) \text{ cm}$	A1	
2(b)(ii)	$k = F / x$	C1	Allow $e / \Delta x / \Delta L$ for x
	$= (0.300 \times 9.81) / 0.119$	C1	ecf of extension from 2b(i) Do not allow $g = 10$
	$= 24.7 \text{ N m}^{-1}$	A1	Allow 25
2(b)(iii)	$0.2 / 11.9$ or $0.005 / 0.300$	C1	ecf of extension from 2b(i) 1.68% and 1.67% imply the C mark
	% uncertainty $= [(0.2 / 11.9) + (0.005 / 0.300)] \times 100$		
	$= 3\%$	A1	
	OR $k_{\text{max}} = (0.305 \times 9.81) / 0.117$ or $k_{\text{min}} = (0.295 \times 9.81) / 0.121$ $= 25.6$ $= 23.9$	(C1)	Allow answers to two or more SF that are correct when rounded by the examiner to 1SF.
	% uncertainty $= [(25.6 - 23.9) / 2] / 24.7$ $= 3\%$	(A1)	allow $(25.6 - 24.7) / 24.7$ and $(24.7 - 23.9) / 24.7$ which lead to 4% and 3% respectively.

6. MAR 26/P22/Q1/b,c

1(a)	rate of change of momentum	B1
1(b)	$\text{kg ms}^{-1} / \text{s} = \text{kg ms}^{-2}$	B1
1(c)	(Base units of) $F = \text{kg ms}^{-2}$ and (Base units of) $m = \text{kg}$ and (Base units of) $A = \text{m}^2$	C1
	(Base units of) RHS $= \text{kg ms}^{-2} \times \text{m}^2 \text{s}^{-1} / (\text{kg} \times \text{m}^2)$ $= \text{ms}^{-3}$ (Base units of) LHS $= \text{ms}^{-1}$ and (base units of) RHS $= \text{ms}^{-3}$ so equation is not homogeneous	A1

7. O/N 25/P21/Q2/b(ii)

2(b)(ii)	SI base units of D : kg m s^{-2}	C1
	base units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1})$ $= \text{kg m}^{-1} \text{s}^{-1}$	A1

8. O/N 25/P22/Q1/a,c

1(a)	air temperature: K and air pressure: $\text{kg m}^{-1} \text{s}^{-2}$	B1
	scalar only ticked for both air temperature and air pressure	B1
1(c)	$5 + 3 + (2 \times 4)$ (= 16%)	C1
	absolute uncertainty in $c = 1.8 \times 0.16$ $= (\pm) 0.3$	A1

9. O/N 25/P24/Q1/a(i)

1(a)(i)	$v_H = 28 \times \cos 34^\circ$ $= 23 \text{ m s}^{-1}$	A1
	$v_V = 28 \times \sin 34^\circ$ $= 16 \text{ m s}^{-1}$	A1

10. O/N 25/P24/Q5/b(iii,iv)

5(b)(iii)	calculation of a fractional or percentage uncertainty in one quantity $0.001 / 0.864$ or $0.002 / 0.496$ or $0.02 / 1.38$ or $0.001 / 0.276$	C1
	percentage uncertainty = $[(0.001 / 0.864) + (2 \times 0.002 / 0.496) + (0.02 / 1.38) + (0.001 / 0.276)] \times 100$ $= 2.7\%$	A1
5(b)(iv)	$\Delta \rho = 0.03 \times 1.12 \times 10^{-6}$ $= 3 \times 10^{-8} \Omega \text{ m}$	A1

11. M/J 25/P22/Q1/a

1(a)	acceleration and displacement identified as vectors (and no others)	B1
	speed, temperature and gravitational potential energy identified as scalars (and no others)	B1

12. M/J 25/P24/Q1/b(ii)

1(b)(ii)	component of weight = $75 \times \cos 42^\circ$ $= 56 \text{ N}$	A1
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13. MAR 25/P22/Q1/a,b(ii,iii)

1(a)	how close the (measured) value is to the true value (of the quantity)	A1
1(b)(ii)	$(0.5 / 31.3)$ or $(0.01 / 1.53)$ % uncertainty = $(0.016 \times 100) + 3 \times (0.0065 \times 100)$	C1
	$= 4\%$	A1

1(b)(iii)	The ranges (of the densities of A and B) overlap or the (calculated) density of A is within the range of the density of B or the difference in densities is within the uncertainty (of B)	M1
	(so) they could be the same	A1

14. O/N 24/P21/Q1/b(ii),c

1(b)(i)	density = $0.243 / (0.0541 \times 0.1109 \times 0.0162)$	C1
	= 2500 kg m^{-3}	A1
1(b)(ii)	(0.001 / 0.243) or (0.01 / 5.41) or (0.01 / 11.09) or (0.01 / 1.62)	C1
	$(\Delta\rho/\rho) = (\Delta m/m) + (\Delta x/x) + (\Delta y/y) + (\Delta z/z)$ = (0.001 / 0.243) + (0.01 / 5.41) + (0.01 / 11.09) + (0.01 / 1.62) (= 0.013)	C1
	percentage uncertainty in ρ = 0.013×100 = 1.3% (allow 1 s.f. answer of 1%)	A1
1(c)	zero error (on calipers / balance) or incorrect calibration (of calipers / balance)	B1

15. O/N 24/P22/Q1/a,b

1(a)	a quantity with magnitude and direction	B1
1(b)(i)	SI base units of D : kg m s^{-2}	C1
	SI base units of r : m and v : m s^{-1}	C1
	base units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1})$ = $\text{kg m}^{-1} \text{ s}^{-1}$	A1
1(b)(ii)	$W = U + D$	A1
1(b)(iii)	$U = 830 \times 9.81 \times 4.6 \times (10^{-2})^3$ (= 0.037 N)	C1
	$W = 0.037 + 0.32$ = 0.36 N	A1

16. M/J 24/P21/Q1/a

1(a)	units of F_D : kg m s^{-2}	M1
	units of ρ : kg m^{-3} and units of A : m^2 and units of v : m s^{-1} or units of v^2 : $\text{m}^2 \text{ s}^{-2}$	M1
	$\text{kg m s}^{-2} = C \text{ kg m s}^{-2}$ and comment '(so) C has no units' / unit terms cancelled or $C = \text{kg m s}^{-2} / (\text{kg m}^{-3} \text{ m}^2 \text{ s}^{-2})$ and comment '(so) C has no units' / unit terms cancelled	A1

17. M/J 24/P21/Q2/b(ii)

2(b)(ii)	$u = 22 \times \sin 40^\circ$ = 14 m s^{-1}	A1
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18. M/J 24/P22/Q1/a

1(a)	charge underlined (and no others)	B1
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19. M/J 24/P23/Q1/a

1(a)	units of F : kg m s^{-2}	C1
	units of r : m and units of v : m s^{-1}	A1
	units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1}) = \text{kg m}^{-1} \text{s}^{-1}$	

20. MAR 24/P22/Q1/a,b

1(a)	current and mass only ticked	A1
1(b)	(power =) work (done) / time	C1
	units of power = J s^{-1} = $\text{kg m}^2 \text{s}^{-2} / \text{s}$ = $\text{kg m}^2 \text{s}^{-3}$	A1

21. O/N 23/P21/Q1/a,b

1(a)	scalar and vector have magnitude	B1
	vector has direction (and scalar does not have direction)	B1
1(b)(i)	$r = [(3 \times 28) / 4\pi]^{1/3}$ = 1.9 cm	A1
1(b)(ii)	percentage uncertainty in $V = (0.5 / 28) \times 100$ (= 1.79%)	C1
	percentage uncertainty in $r = 1.79 / 3$ = 0.6%	A1

22. O/N 23/P22/Q1/a

1(a)	only electric current and time underlined	B1
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23. O/N 23/P23/Q1/a,b(i,iii)

1(a)	charge and power only ticked	B1
1(b)(i)	%D = 0.4% and %L = 0.6%	A1
1(b)(iii)	percentage uncertainty = $0.4 + (2 \times 0.4) + 0.6$	C1
	= 1.8%	A1

24. O/N 23/P23/Q2/a(ii)

2(a)(ii)	$(u_v) = 9.5 \tan 38^\circ$ or $9.5 / \tan 52^\circ$ or $9.5 = u \cos 38^\circ$ and $u_v = u \sin 38^\circ$ or $9.5 = u \cos 38^\circ$ and $u_v = (u^2 - 9.5^2)^{1/2}$	C1
	$u_v = 7.4 \text{ m s}^{-1}$	A1

25. M/J 23/P22/Q1

- (a) (i) force / area (normal to the force) [1]
 (ii) ($p = F/A$ so units are) $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$ [1]
- (b) unit of R : m and unit of t : s and unit of L : m
 unit of ρ : kg m^{-3}
 or
 $\rho = m/V$
 base units of k : $(\text{kg m}^{-1} \text{s}^{-2} \times \text{m}^4 \times \text{kg m}^{-3} \times \text{s}) / (\text{kg} \times \text{m}) = \text{kg m}^{-1} \text{s}^{-1}$ [3]
- (c) R contributes $4 \times 2\%$ or 8% (and L contributes 2%) so R contributes more (to the percentage uncertainty in k) [1]

26. M/J 23/P21/Q1

- (a) (i) work done per unit time 1
 (ii) ($P = W/t$ gives) units: $\text{kg m}^2 \text{s}^{-2} / \text{s} = \text{kg m}^2 \text{s}^{-3}$ 1
- (b) ($I = P/A$ so) units of I : $\text{kg m}^2 \text{s}^{-3} / \text{m}^2$ or kg s^{-3}
 units of f : s^{-1}
 and units of A : m
 and units of v : m s^{-1}
 units of k : $\text{kg s}^{-3} / [(\text{s}^{-1})^2 \text{m}^2 \text{m s}^{-1}]$
 $= \text{kg m}^{-3}$ 3

27. MAR 23/P22/Q1

- (a) only ampere and kelvin underlined 1
- (b) initial speed / velocity is zero
 (non-zero magnitude of) acceleration is constant / uniform (and in a straight line) 2
- (c) (i) $a = 2.75^2 / (2 \times 3.89)$
 $= 0.97 \text{ m s}^{-2}$ 1
- (ii) percentage uncertainty $= (2 \times 0.8) + 0.5$
 $= 2.1\%$ 2
- (iii) absolute uncertainty $= (2.1 / 100) \times 0.97$
 $= 0.02 \text{ m s}^{-2}$ 1

28. MAR 22/P22/Q1

- (a) Fv : kg m s^{-2}
 k : $\text{kg m s}^{-2} / \text{m} \times \text{m s}^{-1}$
 $= \text{kg m}^{-1} \text{s}^{-1}$ [2]
- (b) $F = \rho g V$
 $V = 4/3 \times \pi \times (2.1 \times 10^{-3})^3$ ($= 3.88 \times 10^{-8} \text{ m}^3$)
 $\rho = 4.8 \times 10^{-4} / 9.81 \times V$
 $= 1300 \text{ kg m}^{-3}$ [3]
- (c) (i) W downwards, U upwards, Fv upwards [1]
- (ii) $F_v = 7.2 \times 10^{-4} - 4.8 \times 10^{-4}$
 $= 2.4 \times 10^{-4} \text{ (N)}$
 velocity $= 2.4 \times 10^{-4} / (17 \times 2.1 \times 10^{-3})$
 $= 6.7 \times 10^{-3} \text{ m s}^{-1}$ [2]

29. O/N 22/P22/Q1

- (a) force $\times$ displacement in the direction of the force 1
- (b) $P = F_s / t$
 $= (\text{kg m s}^{-2} \times \text{m}) / \text{s}$
 $= \text{kg m}^2 \text{s}^{-3}$ 2
- (c) (i) $84 \times 10^3 = v^3 \times 0.56$
 $v = 53 \text{ m s}^{-1}$ 2
- (ii) percentage uncertainty $= (5\% + 7\%) / 3 (= 4\%)$
or
fractional uncertainty $= (0.05 + 0.07) / 3 (= 0.04)$
absolute uncertainty $= 0.04 \times 53$
 $= (\pm) 2 \text{ m s}^{-1}$ 2

30. O/N 22/P21/Q1

- (a)
- | | |
|------------------|----------|
| base quantity | coulomb |
| base unit | current |
| derived quantity | force |
| derived unit | kilogram |
- any two joined correctly
all four joined correctly 2
- (b) (i) the measurements have a small range 1
(ii) (average of the) measurements not close to the true value 1
- (c) (i) percentage uncertainty $= (3 + 5 + 4) / 2$
 $= 6\%$ 2
- (ii) absolute uncertainty $= (6 / 100) \times 15.0$
 $= 0.9 \text{ ms}^{-1}$ 1

31. O/N 22/P23/Q1

- (a) (SI base unit of α) $= \text{m}^3 \text{s}^{-1} \times \text{m} / \text{m}^4 = \text{s}^{-1}$ 1
- (b) (percentage uncertainty) $= 3 + 4 + 2 \times 4 = 15 (\%)$ 1
- (c) $\alpha = QL / r^4$
 $= 2.72 \times 10^{-8} \times 2.5 \times 10^{-2} / (7.1 \times 10^{-5})^4$
 $= 2.7 \times 10^7$
absolute uncertainty $= 0.15 \times [2.7 \times 10^7]$
 $= 0.4 \times 10^7$
 $\alpha = (2.7 \pm 0.4) \times 10^7 \text{ s}^{-1}$ 3

32. M/J 22/P22/Q1

- (a) only ampere and kilogram underlined 1
- (b) (i) percentage uncertainty $= 2.1 + 0.6 + (1.5 \times 2)$
 $= 5.7\%$ 2
- (ii) absolute uncertainty $= (5.7 / 100) \times 8.245 \times 10^9$
 $(= 4.7 \times 10^8 \text{ Pa or } 0.47 \times 10^9 \text{ Pa})$
 $E = (8.2 \pm 0.5) \times 10^9 \text{ Pa}$ 2

33. M/J 22/P23/Q1

- (a) $\rho = m / V$
 $V = (4/3) \times \pi \times r^3$
 $= (4/3) \times \pi \times (3.42/2)^3$
 $(= 20.9 \text{ cm}^3)$
 $\rho = 67 / 20.9$
 $= 3.2 \text{ g cm}^{-3}$ 3
- (b) $\% \rho = \%m + 3 \times \%d$
 $= [(2/67) \times 100] + [3 \times (0.02/3.42) \times 100]$
 $= 3.0\% + 3 \times 0.58\%$
 $= 4.7\% \text{ or } 5\%$ 2

34. O/N 21/P23/Q2

- (a) force $\times$ displacement in the direction of the force [1]
- (b) units: $\text{kg m s}^{-2} \times \text{m} = \text{kg m}^2 \text{s}^{-2}$ [1]
- (c) T_1 : K and T_2 : K
 A : m^2 and t : s and L : m
 $c = (\text{kg m}^2 \text{s}^{-2} \text{m}) / (\text{m}^2 \text{K s})$
 $= \text{kg m s}^{-3} \text{K}^{-1}$ [3]

35. O/N 21/P22/Q1

- (a) 10^{12} 2
pico (p) 1
- (b) ampere and metre both underlined (and no other units underlined) 1
- (c) (i) percentage uncertainty $= 3.5 + (3.0 \times 2) + 2.5 + 2.0$
 $= 14\%$ 2
- (ii) absolute uncertainty $= 4.1 \times 10^{-7} \times 14 / 100$
 $= 6 \times 10^{-8} \Omega \text{m}$ 1

36. O/N 21/P21/Q1

- (a) mass / volume 1
- (b) (i) (vernier/digital) calipers 1
- (ii) percentage uncertainty $= (0.0004 / 0.0420) \times 100$
 $= 1\%$ 1
- (c) (i) $\text{kg m}^{-3} = \text{kg} \times \text{m}^n / \text{m}$ or $\text{kg m}^{-3} = \text{kg} \times \text{m}^n \times \text{m}^{-1}$
 $-3 = n - 1$ and (so) $n = -2$ 2
- (ii) $(\Delta\rho/\rho) = (\Delta M/M) + 2(\Delta r/r) + (\Delta L/L)$
percentage uncertainty $= [(0.001/1.072) + 2 \times (0.0004/0.0420) + (0.0001/0.1242)] (\times 100)$
 $= 0.09\% + 2 \times 0.95\% + 0.08\%$
 $= 2\%$ 3
- (iii) $\rho = (1.072 \times 0.0420^{-2}) / (2.094 \times 0.1242)$
 $= 2337 \text{ (kg m}^{-3}\text{)}$
 $\Delta\rho = 0.021 \times 2337$
 $= 49 \text{ (kg m}^{-3}\text{)}$
 $\rho = (2340 \pm 50) \text{ kg m}^{-3}$ 3

37. M/J 21/P21/Q1

- (a) mass / volume 1
- (b) (i) absolute uncertainty $= 4.0 \times (5 / 100)$
 $= (\pm) 0.2 \text{ cm}$ 1
- (ii) percentage uncertainty $= 2 + 4 + (5 \times 2)$
 $= (\pm) 16\%$ 2
- (c) $p = F / A$ or $p = W / A$
 $p = (19.5 \times 10^{-3} \times 9.81) / (4.0 \times 10^{-2})^2$
 $= 120 \text{ Pa}$ 3

38. O/N 20/P23/Q1

- (a) (i) wavelength $= 8.5 \times 10^{-5} \text{ m}$ [1]
- (ii) $f = v / \lambda$ or c / λ
 $= 3.0 \times 10^8 / 8.5 \times 10^{-5} (= 3.5 \times 10^{12})$
 $= 3.5 \text{ THz}$ [2]
- (iii) infrared [1]
- (b) (implied) percentage uncertainty in $I = 4\%$
or
(implied) fractional uncertainty in $I = 0.04$
percentage uncertainty in $E = 5\% + (4\% \times 2) = 13\%$ [2]

39. M/J 20/P22/Q1

- (a) (velocity =) change in displacement / time (taken) [1]
- (b) units of F : kg m s^{-2}
units of k : $\text{kg m s}^{-2} / [\text{m}^2 \times (\text{m s}^{-1})^2]$
 $= \text{kg m}^{-3}$ [2]
- (c) $P = Fv$
 $4.8 \times 10^4 = 0.24 \times 5.1 \times v^3$
 $v = 34 \text{ m s}^{-1}$ [3]

40. M/J 20/P21/Q1

- (a) (work =) force $\times$ displacement
units: $\text{kg m s}^{-2} \times \text{m} = \text{kg m}^2 \text{ s}^{-2}$ [2]
- (b) (i) units of Q : As
units of C : $\text{kg}^{-1} \text{ m}^{-2} \text{ A}^2 \text{ s}^4$ [2]
- (ii) 1. e.g. reading scale from different angles
(wrongly) interpolating between scale readings/divisions
2. e.g. zero error
wrongly calibrated scale [2]

41. MAR 20/P22/Q1

- (a) time
(electric) current
allow amount of substance
allow luminous intensity
any two of the above quantities, 1 mark each [2]
- (b) (i) $g = (4\pi^2 \times 1.50) / (2.48^2)$
 $= 9.63 \text{ m s}^{-2}$ [1]
- (ii) percentage uncertainty = $2 + (3 \times 2)$
or fraction uncertainty = $0.02 + (0.03 \times 2)$
percentage uncertainty = 8% [2]
- (iii) absolute uncertainty = 0.08×9.6
 $= 0.8 \text{ m s}^{-2}$ [1]

42. M/J 20/P23/Q1

- (a) similarity: both have magnitude
difference: distance is a scalar/does not have direction
or
displacement is a vector/has direction 2
- (b) (i) the measurements have a small range 1
(ii) the (average of the) measurements is not close to the true value 1

43. O/N 19/P22/Q1

- (a) scalar quantity has (only) magnitude
vector quantity has magnitude and direction [2]
- (b) (i) $E = F / Q$
 $= \text{kg m s}^{-2} / \text{A s} = \text{kg m A}^{-1} \text{s}^{-3}$ [2]
- (ii) $b = Q / x^2 E$
 $= \text{A s} / \text{m}^2 \text{kg m A}^{-1} \text{s}^{-3}$
 $= \text{A}^2 \text{s}^4 \text{kg}^{-1} \text{m}^{-3}$ [2]

44. O/N 19/P21/Q1

- (a) (i) mass in range 1–20 g [1]
(ii) wavelength in range $1 \times 10^{-8} \text{ m}$ to $4 \times 10^{-7} \text{ m}$ [1]
- (b) (i) $T = 2\pi \times (200 \times 10^{-3} / 25)^{0.5}$
 $= 0.56 \text{ s}$ [1]
- (ii) percentage uncertainty = $(2\% + 8\%) / 2$ (= 5%)
or
fractional uncertainty = $(0.02 + 0.08) / 2$ (= 0.05)
 $\Delta T = 0.56 \times 0.05$
 $= 0.028 \text{ (s)}$
 $T = (0.56 \pm 0.03) \text{ s}$ [3]

45. M/J 19/P21/Q1

(a) (velocity =) change in displacement / time (taken) [1]

(b) (i) $k = [1.29 \times (3.3 \times 10^2)^2] / 9.9 \times 10^4$
 $= 1.4$ [2]

(ii) percentage uncertainty = $(3 \times 2) + 4 + 2$ (= 12%)

or

fractional uncertainty = $(0.03 \times 2) + 0.04 + 0.02$ (= 0.12)

$\Delta k = 0.12 \times 1.42$

$= 0.17$ (allow to 1 significant figure)

$k = 1.4 \pm 0.2$ [3]

46. M/J 19/P23/Q1

(a) (i) potential difference / current [1]

(ii) $R = 4.0 \times 10^9 (\Omega)$

$I = 0.60 / 4.0 \times 10^9 = 1.5 \times 10^{-10} (\text{A})$

$I = 150 \text{ pA}$ [2]

(b) units of energy: $\text{kg m}^2 \text{s}^{-2}$

units of charge: A s

units of potential difference: $(\text{kg m}^2 \text{s}^{-2} / \text{A s}) = \text{kg m}^2 \text{A}^{-1} \text{s}^{-3}$ [3]

47. MAR 19/P22/Q1

(a) kilogram / kg

kelvin / K [2]

(b) units for v : m s^{-1} and units for F : kg m s^{-2}

units for e : A s

units for μ : $\text{m s}^{-1} \text{A s} / \text{kg m s}^{-2}$

$= \text{A kg}^{-1} \text{s}^2$ [3]

48. M/J 19/P22/Q1

(a) absolute uncertainty = $(1.6 / 100) \times 0.0125$

$= 2 \times 10^{-4} \text{ m}$

1

(b) (i) $\rho = (4 \times 0.38) / (\pi \times 0.0125^2)$

$= 3100 \text{ N m}^{-2}$

1

(ii) percentage uncertainty = $2.8 + (2 \times 1.6)$ (= 6%)

or

fractional uncertainty = $0.028 + (2 \times 0.016)$ (= 0.06)

absolute uncertainty = 0.06×3100

$= 190 \text{ N m}^{-2}$ (allow to 1 significant figure) 2

49. M/J 18/P22/Q1

(a) rate of change of momentum [1]

(b) kg m s^{-2} [1]

(c) units for Q : A s and for r : m

units for $\epsilon = (\text{A s} \times \text{A s}) / (\text{kg m s}^{-2} \times \text{m}^2)$

$= \text{A}^2 \text{kg}^{-1} \text{m}^{-3} \text{s}^4$ [2]

50. M/J 18/P21/Q1

- (a) a scalar has magnitude (only)
a vector has magnitude and direction [2]
- (b) power: scalar
temperature: scalar
momentum: vector
(two correct 1 mark, all three correct 2 marks) [2]
- (c) (i) arrow labelled R in a direction from 5° to 20° north of west [1]
(ii) $v^2 = 28^2 + 95^2 - (2 \times 28 \times 95 \times \cos 115^\circ)$
or
 $v^2 = [(95 + 28 \cos 65^\circ)^2 + (28 \sin 65^\circ)^2]$
 $v = 110 \text{ ms}^{-1}$ (109.8 ms^{-1})
or (scale diagram method)
triangle of velocities drawn
 $v = 110 \text{ ms}^{-1}$ (allow $108\text{--}112 \text{ ms}^{-1}$) [2]

51. M/J 18/P23/Q1

- (a) (i) zero error or wrongly calibrated scale 1
(ii) reading scale from different angles or wrongly interpolating between scale readings/divisions 1
- (b) (i) $P = V^2/R$ or $P = VI$ and $V = IR$
 $P = 5.0^2 / 125$ or 5.0×0.04 or $(0.04)^2 \times 125$
 $= 0.20 \text{ W}$ 2
(ii) $\%V = 2\%$ or $\Delta V / V = 0.02$
 $\%P = (2 \times 2\%) + 3\%$ or $\%P = (2 \times 0.02 + 0.03) \times 100$
 $= 7\%$ 2
(iii) absolute uncertainty in $P = (7/100) \times 0.20$
 $= 0.014$
power $= 0.20 \pm 0.01 \text{ W}$ or $(2.0 \pm 0.1) \times 10^{-1} \text{ W}$ 2

52. O/N 17/P21/Q1

- (a) units of F : kg ms^{-2}
units of ρ : kg m^{-3} and units of v : ms^{-1}
units of K : $\text{kg ms}^{-2} / [\text{kg m}^{-3} (\text{ms}^{-1})^2]$
 $= \text{m}^2$ [3]
- (b) (i) $K\rho = 1.5 / 33^2$
 $= 1.38 \times 10^{-3}$
 $F_D = 1.38 \times 10^{-3} \times 25^2$ or $F_D / 1.5 = 25^2 / 33^2$
 $F_D = 0.86 \text{ N}$ [2]
(ii) $a = (1.5 - 0.86) / (1.5 / 9.81)$ or $a = 9.81 - [0.86 / (1.5 / 9.81)]$
 $a = 4.2 \text{ ms}^{-2}$ [2]
- (c) initial acceleration is $g/9.81$ (ms^{-2})/acceleration of free fall
acceleration decreases
final acceleration is zero [3]

53. O/N 17/P23/Q1

- (a) (i) work (done)/time (taken) **or** energy (transferred)/time (taken) [1]
 (ii) Correct substitution of base units of all quantities into any correct equation for power.
 Examples:
 $(P = E/t \text{ or } W/t \text{ gives}) \text{ kg m}^2 \text{ s}^{-2} / \text{s} = \text{kg m}^2 \text{ s}^{-3}$
 $(P = Fs/t \text{ or } mgh/t \text{ gives}) \text{ kg m s}^{-2} \text{ m} / \text{s} = \text{kg m}^2 \text{ s}^{-3}$
 $(P = \frac{1}{2}mv^2/t \text{ gives}) \text{ kg (m s}^{-1})^2 / \text{s} = \text{kg m}^2 \text{ s}^{-3}$
 $(P = Fv \text{ gives}) \text{ kg m s}^{-2} \text{ m s}^{-1} = \text{kg m}^2 \text{ s}^{-3}$
 $(P = VI \text{ gives}) \text{ kg m}^2 \text{ s}^{-2} \text{ A}^{-1} \text{ s}^{-1} \text{ A} = \text{kg m}^2 \text{ s}^{-3}$ [1]
- (b) (i) units of A : m^2 **and** units of T : K
 units of k : $\text{kg m}^2 \text{ s}^{-3} / \text{m}^2 \text{ K}^4$
 $= \text{kg s}^{-3} \text{ K}^{-4}$ [2]
 (ii) curve from the origin with increasing gradient [1]

54. M/J 17/P22/Q1

- (a) kelvin, mole, ampere, candela
 any two [1]
- (b) use of resistivity = RA/l **and** $V = IR$ (to give $\rho = VA/I l$)
 units of V : (work done / charge) $\text{kg m}^2 \text{ s}^{-2} (\text{A s})^{-1}$
 units of resistivity: $(\text{kg m}^2 \text{ s}^{-3} \text{ A}^{-1} \text{ A}^{-1} \text{ m})$
 $= \text{kg m}^3 \text{ s}^{-3} \text{ A}^{-2}$
or
 use of $R = \rho L/A$ **and** $P = I^2 R$ (gives $\rho = PA/I^2 L$)
 units of P : $\text{kg m}^2 \text{ s}^{-3}$
 units of resistivity: $(\text{kg m}^2 \text{ s}^{-3} \times \text{m}^2) / (\text{A}^2 \times \text{m})$
 $= \text{kg m}^3 \text{ s}^{-3} \text{ A}^{-2}$ [3]
- (c) (i) $\rho = (RA/l)$
 $= (0.03 \times 1.5 \times 10^{-6}) / 2.5 (= 1.8 \times 10^{-8})$
 $= 18 \text{ n}\Omega \text{ m}$ [3]
- (ii) 1. precision is determined by the range in the measurements/values/readings/data/results
 2. metre rule measures to $\pm 1 \text{ mm}$ and micrometer to $\pm 0.01 \text{ mm}$ (so there is less (percentage) uncertainty/random error) [2]

55. M/J 17/P21/Q1

- (a) (stress =) force / area **or** $\text{kg m s}^{-2} / \text{m}^2$
 $= \text{kg m}^{-1} \text{ s}^{-2}$ [2]
- (b) (i) $0.58 = 2\pi \times [(4 \times 0.500 \times 0.600^3) / (E \times 0.0300 \times 0.00500^3)]^{0.5}$
 $E = [4\pi^2 \times 4 \times 0.500 \times (0.600)^3] / [(0.58)^2 \times 0.0300 \times (0.00500)^3]$
 $= 1.35 \times 10^{10} \text{ (Pa)}$
 $= 14 \text{ (13.5) GPa}$ [3]
- (ii) 1. (accuracy determined by) the closeness of the value(s)/measurement(s) to the true value (precision determined by) the range of the values/measurements [2]
 2. l is (cubed so) $3 \times$ (percentage/fractional) uncertainty
and T is (squared so) $2 \times$ (percentage / fractional) uncertainty
and (so) l contributes more [1]

56. M/J 17/P23/Q1

- (a) (i) $R = 7(.0) \text{ N}$ [1]
 (ii) $R = 13 \text{ N}$ [1]
- (b) (i) forces resolved: $18 \sin 65^\circ$ (vertical) **and** $55 + 18 \cos 65^\circ$ (horizontal)
 or
 scale drawing: correct triangle drawn for forces
 $F = [(18 \sin 65^\circ)^2 + (55 + 18 \cos 65^\circ)^2]^{1/2} = 65 \text{ (64.7) N}$
 or
 scale drawing: scale given, length of resultant given correctly, $\pm 1 \text{ N}$ [2]
- (ii) angle = $\tan^{-1} [18 \sin 65^\circ / (55 + 18 \cos 65^\circ)] = \tan^{-1} (16.3 / 62.6)$
 or
 scale drawing: correct angle measured/direction correct on diagram below the 55 N force
 angle = 15 (14.6)° (below the 55 N force)
 or
 scale drawing: angle = $15^\circ \pm 1^\circ$ [2]
- (c) (resultant) force = mass $\times$ acceleration
 $80 - 65 = 2.7a$
 $a = 5.6 \text{ m s}^{-2}$ [5.7 if 64.7 N used from (i)] [3]

57. O/N 17/P22/Q1

- (a) (i) micrometer (screw gauge)/digital calipers 1
 (ii) take several readings (and average)
 along the wire **or** around the circumference 2
- (b) (i) $\sigma = 4 \times 25 / [\pi \times (0.40 \times 10^{-3})^2] = 1.99 \times 10^8 \text{ N m}^{-2}$
 or
 $\sigma = 25 / [\pi \times (0.20 \times 10^{-3})^2] = 1.99 \times 10^8 \text{ N m}^{-2}$ 1
- (ii) $\%F = 2\%$ **and** $\%d = 5\%$
 or 2
 $\Delta F / F = \frac{0.5}{25}$ **and** $\Delta d / d = \frac{0.02}{0.4}$
 $\%\sigma = 2\% + (2 \times 5\%)$
 or
 $\%\sigma = [0.02 + (2 \times 0.05)] \times 100$
 $\%\sigma = 12\%$
- (iii) absolute uncertainty = $(12/100) \times 1.99 \times 10^8$
 $= 2.4 \times 10^7$
 $\sigma = 2.0 \times 10^8 \pm 0.2 \times 10^8 \text{ N m}^{-2}$ **or** $2.0 \pm 0.2 \times 10^8 \text{ N m}^{-2}$ 2

58. O/N 16/P22/Q1

- (a) (i) force / area (normal to the force) [1]
 (ii) ($p = F/A$ so) units: $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{ s}^{-2}$
 allow use of other correct equations:
 e.g. ($\Delta p = \rho g \Delta h$ so) $\text{kg m}^{-3} \text{ m s}^{-2} \text{ m} = \text{kg m}^{-1} \text{ s}^{-2}$
 e.g. ($p = W / \Delta V$ so) $\text{kg m s}^{-2} \text{ m} / \text{m}^3 = \text{kg m}^{-1} \text{ s}^{-2}$ [1]
- (b) units for m : kg, t : s and ρ : kg m^{-3}
 units of C : kg/s ($\text{kg m}^{-3} \text{ kg m}^{-1} \text{ s}^{-2}$)^{1/2}
 or
 units of C^2 : kg^2 / s^2 $\text{kg m}^{-3} \text{ kg m}^{-1} \text{ s}^{-2}$
 units of C : m^2 [3]

59. M/J 16/P23/Q1

- (a) scalars: energy, power and time
vectors: momentum and weight [2]
- (b) (i) triangle with right angles between 120 m and 80 m, arrows in correct direction and result displacement from start to finish arrow in correct direction and labelled R [1]
- (ii) 1. average speed $(= 200/27) = 7.4 \text{ m s}^{-1}$ [1]
2. resultant displacement $(= [120^2 + 80^2]^{1/2}) = 144 \text{ (m)}$
average velocity $(= 144/27) = 5.3(3) \text{ m s}^{-1}$
direction $(= \tan^{-1} 80/120) = 34^\circ (33.7)$ [3]

60. O/N 16/P21/Q1

- (a) (density =) mass / volume [1]
- (b) (i) $d = [(6 \times 7.5)/(\pi \times 8100)]^{1/3}$
 $= 0.12(1) \text{ m}$ [1]
- (ii) percentage uncertainty $= (4 + 5)/3$ ($= 3\%$)
or
fractional uncertainty $= (0.04 + 0.05)/3$ ($= 0.03$)
absolute uncertainty $(= 0.03 \times 0.121) = 0.0036$
 $d = 0.121 \pm 0.004 \text{ m}$ [3]

61. M/J 16/P21/Q1

- (a) (i) $(50 \text{ to } 200) \times 10^{-3} \text{ kg}$ or $(0.05 \text{ to } 0.2) \text{ kg}$ [1]
(ii) $(50 \text{ to } 300) \text{ cm}^3$ [1]
- (b) density = mass / volume or $\rho = M/V$
 $V = [\pi(0.38 \times 10^{-3})^2 \times 25.0 \times 10^{-2}]/4$ ($= 2.835 \times 10^{-8} \text{ m}^3$)
 $\rho = (0.225 \times 10^{-3})/2.835 \times 10^{-8}$
 $= 7940 \text{ (kg m}^{-3}\text{)}$
 $\Delta\rho/\rho = 2(0.01/0.38) + (0.1/25.0) + (0.001/0.225)$ [$= 0.061$]
or
 $\%\rho = 5.3\% + 0.40\% + 0.44\%$ ($= 6.1\%$)
 $\Delta\rho = 0.061 \times 7940 = 480 \text{ (kg m}^{-3}\text{)}$
density $= (7.9 \pm 0.5) \times 10^3 \text{ kg m}^{-3}$ or $(7900 \pm 500) \text{ kg m}^{-3}$ [5]

62. M/J 16/P23/Q2

- (a) systematic: the reading is larger or smaller than (or varying from) the true reading by a constant amount
random: scatter in readings about the true reading [2]
- (b) precision: the size of the smallest division (on the measuring instrument)
or
0.01 mm for the micrometer
accuracy: how close (diameter) value is to the true (diameter) value [2]

63. O/N 15/P22/Q1

- (a) $v = f\lambda$
 $\lambda = (3.0 \times 10^8) / (4.6 \times 10^{20})$
 $(= 6.52 \times 10^{-13} =) 0.65(2) \text{ pm}$ [3]
- (b) $t = (8.5 \times 10^{16}) / (3.0 \times 10^8)$
 $(= 2.83 \times 10^8 =) 0.28(3) \text{ Gs}$ [2]
- (c) mass, power and temperature all underlined and no others [1]
- (d) (i) arrow in the direction 30° to 40° south of east [1]
 (ii) triangle of velocities completed (i.e. correct scale diagram) or correct working given
 e.g. $[14^2 + 8.0^2 - 2(14)(8.0) \cos 60^\circ]^{1/2}$
 or $[(14 - 8.0 \cos 60^\circ)^2 + (8.0 \sin 60^\circ)^2]^{1/2}$
 resultant velocity = $12(.2)$ (or 12.0 to 12.4 from scale diagram) m s^{-1} [2]

64. O/N 15/P21/Q1

- (a) temperature
 current
(allow amount of substance, luminous intensity) [2]
- (b) (i) 1. $E = (\text{stress} / \text{strain}) = [\text{force} / \text{area}] / [\text{extension} / \text{original length}]$
 units of stress: $\text{kg m s}^{-2} / \text{m}^2$ and no units for strain
 units of E : $\text{kg m}^{-1} \text{s}^{-2}$ [1]
 2. units for T : s, l : m and M : kg
 $K^2 = T^2 E / M l^3$ hence units: $\text{s}^2 \text{kg m}^{-1} \text{s}^{-2} / \text{kg}^3 (= \text{m}^{-4})$
 units of K : m^{-2} [2]
 (ii) % uncertainty in $E = 4\%$ (for T^2) + 0.6% (for l^3) + 0.1% (for M) + 3% (for K^2)
 $= 7.7\%$
 $E = [(1.48 \times 10^5)^2 \times 0.2068 \times (0.892)^3] / (0.45)^2$
 $= 1.588 \times 10^{10}$
 7.7% of $E = 1.22 \times 10^9$
 $E = (1.6 \pm 0.1) \times 10^{10} \text{ kg m}^{-1} \text{s}^{-2}$ [4]

65. O/N 15/P23/Q1

- (a) energy or W : $\text{kg m}^2 \text{s}^{-2}$
 or
 power or P : $\text{kg m}^2 \text{s}^{-3}$
 intensity or I : $\text{kg m}^2 \text{s}^{-2} \text{m}^{-2} \text{s}^{-1}$ (from use of energy expression)
 or
 $\text{kg m}^2 \text{s}^{-3} \text{m}^{-2}$ (from use of power expression)
 indication of simplification to kg s^{-3} [2]
- (b) (i) ρ : kg m^{-3} , c : ms^{-1} , f : s^{-1} , x_0 : m
 substitution of terms in an appropriate equation and simplification to show K has no units [2]
 (ii) $I = 20 \times 1.2 \times 330 \times (260)^2 \times (0.24 \times 10^{-9})^2$
 $= 3.1 \times 10^{-11} (\text{W m}^{-2})$
 $= 31 (30.8) \text{ pW m}^{-2}$ [3]

66. M/J 15/P22/Q1

- (a) (work =) force $\times$ distance or force $\times$ displacement or ($W =$) $F \times d$
 units of work: $\text{kg m s}^{-2} \times \text{m} = \text{kg m}^2 \text{s}^{-2}$ [2]
- (b) (p.d. =) $\frac{\text{work (done) or energy (transformed) (from electrical to other forms)}}{\text{charge}}$ [1]
- (c) $R = V/I$
 units of V : $\text{kg m}^2 \text{s}^{-2} / \text{A s}$ and units of I : A
 or
 $R = P/I^2$ [or $P = VI$ and $V = IR$]
 units of P : $\text{kg m}^2 \text{s}^{-3}$ and units of I : A
 or
 $R = V^2/P$
 units of V : $\text{kg m}^2 \text{s}^{-2} / \text{A s}$ and units of P : $\text{kg m}^2 \text{s}^{-3}$
 units of R : $(\text{kg m}^2 \text{s}^{-2} / \text{A}^2 \text{s}) = \text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$ [3]

67. M/J 15/P21/Q1

- (a) power = work/time or energy/time or (force $\times$ distance)/time
 $= \text{kg m s}^{-2} \times \text{m s}^{-1} = \text{kg m}^2 \text{s}^{-3}$ [2]
- (b) power = VI [or V^2/R and $V = IR$ or $I^2 R$ and $V = IR$]
 (units of V): $\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$ [2]

68. M/J 15/P23/Q1

- (a) 150 or $1.5 \times 10^2 \text{ Gm}$ [1]
- (b) distance = $2 \times (42.3 - 6.38) \times 10^6 (= 7.184 \times 10^7 \text{ m})$
 (time =) $7.184 \times 10^7 / (3.0 \times 10^8) = 0.24 (0.239) \text{ s}$ [2]
- (c) units of pressure P : $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$
 units of density ρ : kg m^{-3} and speed v : m s^{-1}
 simplification for units of C : $C = v^2 \rho / P$ units: $(\text{m}^2 \text{s}^{-2} \text{kg m}^{-3}) / \text{kg m}^{-1} \text{s}^{-2}$
 and cancelling to give no units for C [3]
- (d) energy and power (*both underlined and no others*) [1]
- (e) (i) vector triangle of correct orientation
 three arrows for the velocities in the correct directions [2]
- (ii) length measured from scale diagram $5.2 \pm 0.2 \text{ cm}$ or components of
 boat speed determined parallel and perpendicular to river flow
 velocity = 2.6 m s^{-1} (allow $\pm 0.1 \text{ m s}^{-1}$) [2]

69. O/N 15/P21/Q2

- (a) $p s = 10^{-12} \text{ (s)}$ or $T = 4 \times 50 \times 10^{-12} \text{ (s)}$
 $v = f\lambda$ or $v = \lambda / T$
 $\lambda = 3.0 \times 10^8 \times 4 \times 50 \times 10^{-12}$
 $= 0.06(0) \text{ m}$ [4]
- (b) $1500 = 3.0 \times 10^8 \times 4 \times \text{time-base setting}$ or $T = 5 \times 10^{-6} \text{ s}$
 time-base setting = $1.3 (1.25) \mu \text{s cm}^{-1}$ [2]

70. O/N 14/P21/Q1

- (a) temperature
current
(allow amount of substance and luminous intensity) [2]
- (b) base units of force constant: $\text{kg ms}^{-2} \text{m}^{-1}$ or kg s^{-2}
base units of time and mass: s and kg
base units of C: $\text{s}(\text{kg s}^{-2} / \text{kg})^{1/2}$ cancelling to show no units [3]

71. O/N 14/P23/Q1

- (a) ampere
kelvin
(allow mole and candela) [2]
- (b) (i) stress: N m^{-2}
 $\text{kg ms}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$ [2]
- (ii) Young modulus = stress/strain and strain has no units
hence units: $\text{kg m}^{-1} \text{s}^{-2}$ [1]

72. M/J 14/P22/Q1

- (a) power = energy/time or work done/time
force: kg ms^{-2} (including from mg in mgh or Fv)
or kinetic energy ($\frac{1}{2}mv^2$): $\text{kg (ms}^{-1})^2$
(distance: m and (time) $^{-1}$: s^{-1}) and hence power: $\text{kg ms}^{-2} \text{ms}^{-1} = \text{kg m}^2 \text{s}^{-3}$ [3]
- (b) Q/t : $\text{kg m}^2 \text{s}^{-3}$
A: m^2 and x: m and T: K
correct substitution into $C = (Qx) / tAT$ or equivalent, or with cancellation
units of C: $\text{kg ms}^{-3} \text{K}^{-1}$ [4]

73. M/J 14/P23/Q1

- (a) current, mass and temperature
two correct 2/2, one omission or error 1/2 [2]
- (b) σ : no units, V : m^3
 E_p : $\text{kg m}^2 \text{s}^{-2}$
C: $\text{kg m}^2 \text{s}^{-2} \times \text{m}^{-3} = \text{kg m}^{-1} \text{s}^{-2}$ [3]

74. O/N 14/P21/Q2

- (a) pressure = force / area (normal to the force) [clear ratio essential] [1]
- (b) (i) $P = mg / A = (5.09 \times 9.81) / A$
 $A = (\pi d^2 / 4) = \pi \times (9.4 \times 10^{-2})^2 / 4 (= 0.00694 \text{ m}^2)$
 $P = 49.93 / 0.00694$
 $= 7200 (7195) \text{ Pa}$ (minimum of 2 s.f. required) [3]
- (ii) $\Delta P / P = \Delta m / m + 2\Delta d / d$
 $= 0.01 / 5.09 + (2 \times 0.1) / 9.4 (= 0.0020 + 0.021 \text{ or } 2.3\%)$
 $\Delta P = 170 (165 \text{ to } 167) \text{ Pa}$ [3]
- (iii) $P = 7200 \pm 200 \text{ Pa}$ [1]

75. O/N 14/P23/Q2

- (a) (i) amplitude scale reading 2.2 (cm)
amplitude = $2.2 \times 2.5 = 5.5$ mV [2]
- (ii) time period scale reading = 3.8 (cm)
time period = $3.8 \times 0.5 \times 10^{-3} = 0.0019$ (s)
frequency $f = 1 / 0.0019 = 530$ (526) Hz [3]
- (iii) uncertainty in reading = ± 0.2 in 3.8 (cm) or 5.3% or 0.2 in 7.6 (cm)
or 2.6% [allow other variations of the distance on the x-axis]
actual uncertainty = 5.3% of 526 = 27.7 or 28 Hz
or 2.6% of 526 = 13 or 14 [2]
- (b) frequency = 530 ± 30 Hz or 530 ± 10 Hz [1]

76. M/J 14/P22/Q2

- (a) $\rho = m / V$
 $V = (\pi d^2 / 4) \times t = 7.67 \times 10^{-7} \text{ m}^3$
 $\rho = (9.6 \times 10^{-3}) / [\pi (22.1/2 \times 10^{-3})^2 \times 2.00 \times 10^{-3}]$
 $\rho = 12513 \text{ kg m}^{-3}$ (allow 2 or more s.f.) [3]
- (b) (i) $\Delta\rho/\rho = \Delta m/m + \Delta t/t + 2\Delta d/d$
 $= 5.21\% + 0.50\% + 0.905\%$ [or correct fractional uncertainties] [3]
 $= 6.6\%$ (6.61%) [1]
- (ii) $\rho = 12500 \pm 800 \text{ kg m}^{-3}$

77. O/N 13/P21/Q1

- (a) kelvin / K
ampere / amp / A
[allow mole / mol and candela / Cd] [2]
- (b) (i) energy OR work = force $\times$ distance [allow any energy expression]
units: $\text{kg m s}^{-2} \times \text{m}$ OR $\text{kg (ms}^{-1})^2$ for $\frac{1}{2}mv^2$ or mc^2
(ignore any numerical factor)
 $= \text{kg m}^2 \text{ s}^{-2}$ [2]
- (ii) units: ρ : kg m^{-3} g : ms^{-2} A : m^2 l_0 : m
 C : $\text{kg m}^2 \text{ s}^{-2} / \text{kg}^2 \text{ m}^{-6} \text{ m}^2 \text{ s}^{-4} \text{ m}^2 \text{ m}^3$ [any subject]
 $= \text{kg}^{-1} \text{ m s}^2$ (allow $\text{m s}^2 / \text{kg}$) [3]

78. O/N 13/P23/Q1

- volume = $\pi (14 \times 10^{-3})^2 \times 12 \times 10^{-3} (= 7.389 \times 10^{-6} \text{ m}^3)$
density = mass / volume [any subject]
mass = $6.8 \times 10^3 \times 7.389 \times 10^{-6} = 0.0502$
weight = mg
 $= 0.0502 \times 9.81 = 0.49 \text{ N}$ (mark not awarded if not to **two** s.f.) [4]

79. O/N 13/P21/Q2

- (a) $d = v \times t$
 $t = 0.2 \times 4$ (allow $t = 0.2 \times 2$)
 $d = 3 \times 10^8 \times 0.8 \times 10^{-6}$ OR $3 \times 10^8 \times 0.4 \times 10^{-6}$
 $d = 240 \text{ m}$ hence distance from source to reflector = 120 m [4]
- (b) speed of sound 300 cf speed of light 3×10^8 OR time = $240 / 300$ (= 0.8)
 OR time = $120 / 300$ (= 0.4)
 sound slower by factor of 10^6 OR time for one division $0.8 / 4$
 OR time for one division $0.4 / 2$
 time base setting 0.2 s cm^{-1} [unit required] [3]

80. O/N 13/P23/Q2

- (a) SI units for T : s, R : m and M : kg (or seen clearly in formula)
 $K = T^2 M / R^3$ units: $\text{s}^2 \text{ kg m}^{-3}$ (allow $\text{s}^2 \text{ kg} / \text{m}^3$ or $\frac{\text{s}^2 \text{ kg}}{\text{m}^3}$) [2]
- (b) % uncertainty in K : 1% (for T) + 3% (for R) + 2% (for M) OR = 6%
 $K = [(86400)^2 \times 6 \times 10^{24}] / (4.23 \times 10^7)^3 = 5.918 \times 10^{11}$
 6% of $K = 0.355 \times 10^{11}$
 $K = (5.9 \pm 0.4) \times 10^{11}$ (SI units) correct power of ten required for both [4]
 [incorrect % value then max. 1]